Tree mortality and ecosystem functional properties

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Coauthor notes (not part of the manuscript). Section order follows Nature Ecology & Evolution: Abstract, Main, Results, Discussion, Methods, Data availability, Code availability, References, Acknowledgements, Funding, Author information, Extended Data / Supplementary Information.

Main findings (AutoGluon, deadtrees v5, 101 sites, 361 site-years)

1. **Knowing about tree mortality makes carbon uptake more predictable, especially in the years when trees died.** Adding mortality lowers the error (MAE) of NEP_{max} by 11–12 % over all site-years and by 29–32 % in high-mortality years; WUE improves mainly where mortality happened (−17 to −22 % in high-mortality years); GPP_{sat} improves less (−4 to −12 %) and ET_{max} not at all (Table 2).
2. **Without mortality, models over-estimate carbon uptake in ecosystems with high mortality; mortality corrects this bias.** At >30 % relative mortality the model without mortality over-predicts NEP_{max} by about 4 and GPP_{sat} by about 4.3 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ and WUE by about 0.8 $\mathrm{g\,C\,mm^{-1}}$. The bias grows with mortality; adding mortality reduces it by about two-thirds, and the correction is significant — for all three learners and robust to resampling whole sites, leaving out any single site and removing the most extreme sites (Fig. 2).
3. **The effect follows the amount of mortality, not the ecosystem type.** Mortality pushes predicted GPP_{sat}, NEP_{max} and WUE down at high-mortality sites (86–91 % of them) and slightly up at low-mortality sites (2–10 %); the direction tracks relative mortality closely ($\rho = -0.78$ to -0.83) and does not depend on tree cover or vegetation class. Relative mortality alone carries most of the information (Figs. 3–5).

In one sentence: tree mortality mapped from remote sensing makes ecosystem carbon uptake more predictable — without it, models over-estimate the productivity of woody ecosystems with high mortality, and the correction scales with the amount of mortality regardless of ecosystem type.

What changed in this version

- **Dataset:** deadtrees v5 mortality maps (delivered 2026-10-06).
- **Main learner: AutoGluon** (AutoML ensemble). XGBoost and LightGBM (Optuna-tuned) are in the Appendix as a check.
- **Main test:** paired bootstrap of the MAE at the **site-year level** (every year is an observation, whole sites resampled; Table 2). Site-level tests and the per-site figures are in the Appendix.
- Only **relative mortality** is shown in the main text; the other mortality metrics are in the Appendix.

Abstract

Draft for Nature Ecology & Evolution: unreferenced, ≤200 words.

Tree mortality is increasing in forests worldwide, yet how it changes ecosystem functioning across sites remains poorly quantified. Here we combine eddy-covariance measurements from 101 sites with at least 30 % tree cover (361 site-years; forests, woody savannas, shrublands and wetlands) with remote-sensing maps of tree mortality and machine learning, and show that knowing about mortality makes their carbon uptake markedly more predictable. Adding mortality to climate and plant traits reduced the prediction error of maximum net ecosystem productivity by 11–12 % overall and by about 30 % in years with high mortality, and improved water-use efficiency mainly where mortality occurred; maximum evapotranspiration did not improve. Without mortality, models overestimated carbon uptake and water-use efficiency in high-mortality years, a bias that grew with the amount of mortality and that mortality information reduced by about two-thirds. The direction of the effect followed the amount of mortality rather than vegetation type or tree cover, and the results held across three learning algorithms and when the climate variables used to calculate each property were left out. Mortality maps can thus correct a systematic over-estimate of carbon uptake in woody ecosystems with high mortality.

Main

- Terrestrial ecosystems regulate global **carbon, water, and energy balances**; forest functioning is closely linked to vegetation structure and functional traits.
- Increasing **drought, heat, and biotic disturbances** are causing widespread tree mortality, with potentially large consequences for ecosystem functioning.
- Mortality alters **canopy structure and leaf area**, affecting carbon uptake, evapotranspiration, and energy partitioning, with cascading effects on soils, nutrients, and vegetation recovery.
- The magnitude and trajectory of these impacts likely depend on **vegetation type and functional trait composition**.
- However, quantitative understanding of mortality impacts across ecosystems remains limited due to **spatial and temporal constraints of field observations** and limited integration of mortality and trait data.
- Advances in **remote sensing, machine learning, citizen science, and trait databases** now enable large-scale assessments of mortality and ecosystem responses.
- This creates an opportunity to quantify how mortality affects ecosystem functioning and identify **vegetation types that are more sensitive or resilient**.

Results

Mortality across 101 woody ecosystems

We combined four ecosystem functional properties — light-saturated gross primary productivity (GPP_{sat}), maximum net ecosystem productivity (NEP_{max}), maximum evapotranspiration (ET_{max}) and water-use efficiency (WUE) — from 101 eddy-covariance sites with at least 30 % tree cover (361 site-years) with annual remote-sensing maps of standing deadwood and tree cover (Fig. 1). Most sites are temperate and boreal forests (49 evergreen needleleaf, 30 deciduous broadleaf and 7 mixed forests); the others are wetlands (6), evergreen broadleaf forests (3), shrublands (5) and a woody savanna. We grouped the sites by their maximum relative mortality, the share of the tree cover that died or was lost in a year: Low (< 10.6 %, 41 sites), Mid (38 sites) and High (> 24.0 %, 22

sites). For each property we trained machine-learning models on climate and plant traits (C+T) and on climate, traits and mortality (C+T+M) and evaluated them with leave-one-site-out cross-validation, so that every prediction is made for a site the model has not seen (AutoGluon; XGBoost and LightGBM as checks).

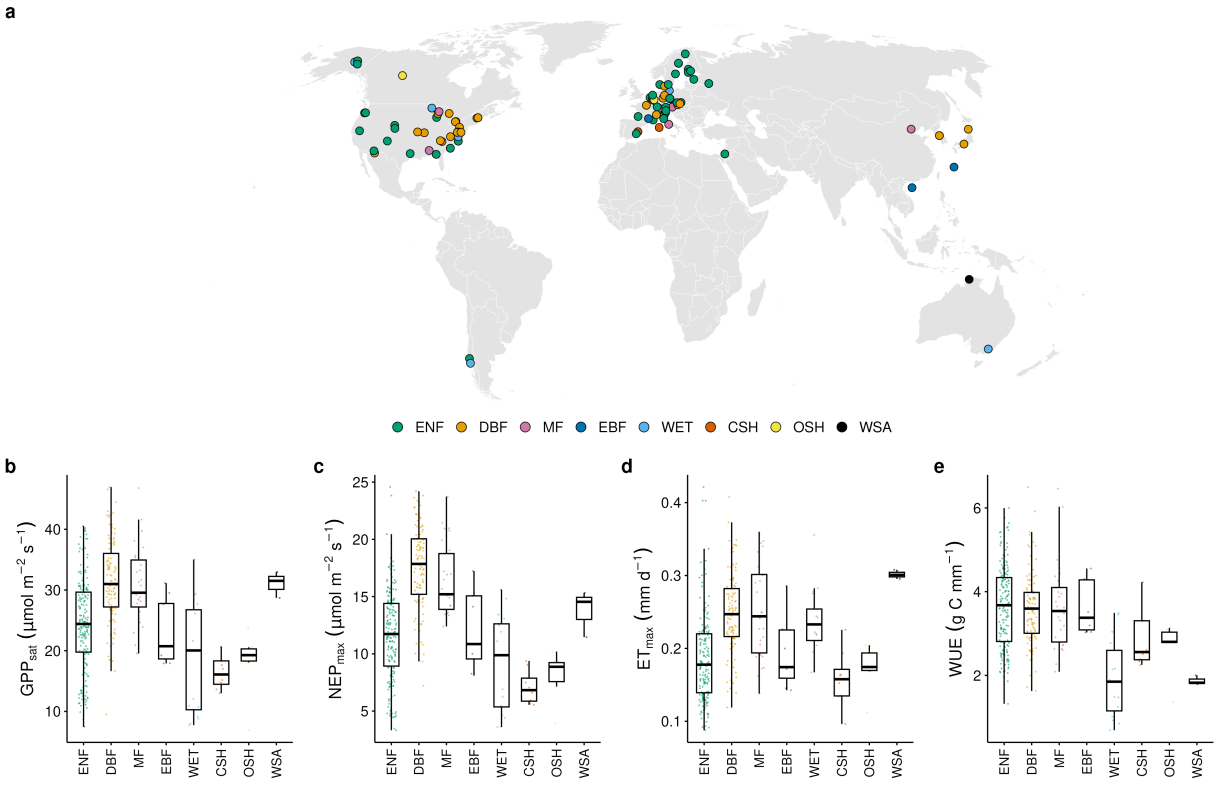


Fig. 1 | The 101 eddy-covariance sites and their ecosystem functional properties. a, Sites with at least 30 % tree cover, coloured by IGBP vegetation class (ENF, evergreen needleleaf forest; DBF, deciduous broadleaf forest; MF, mixed forest; EBF, evergreen broadleaf forest; WET, wetland; CSH, closed shrubland; OSH, open shrubland; WSA, woody savanna); Robinson projection. **b–e**, GPP_{sat} (**b**), NEP_{max} (**c**), ET_{max} (**d**) and WUE (**e**) per vegetation class; points are site-years ($n = 361$), boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range.

Mortality improves predictions

To test whether knowing about mortality makes ecosystem function more predictable, we compared the errors of the two models. Adding mortality raised the mean cross-validated R^2 from 0.35 to 0.40 with the 12-month predictor window and from 0.35 to 0.41 with the 24-month window (Table 1). The gain was largest for NEP_{max} (+0.11), followed by WUE (+0.06) and GPP_{sat} (+0.04 to +0.05); ET_{max} did not improve. With every site-year as an observation and whole sites resampled, mortality reduced the mean absolute error (MAE) of NEP_{max} by 11.8 % (95 % bootstrap interval 6.3–17.0 %) and 10.8 % (5.1–16.0 %) for the 12- and 24-month windows (Table 2). The reduction was largest in the years in which trees died: in site-years with more than 24 % relative mortality the NEP_{max} error fell by 31.6 % (16.0–43.2 %) and 29.2 % (8.6–42.9 %), and the WUE error by 16.9 % (5.1–25.2 %) and 21.8 % (8.2–34.5 %), although WUE did not improve significantly across all site-years (−4.3 and −4.7 %). GPP_{sat} improved less (−4 to −12 %; significant across all site-years at 24 months and in the years with more than 10.6 % mortality), and ET_{max} improved only in the years with more than 10.6 % mortality at 24 months (−5.4 %). Mortality information therefore matters most for the carbon fluxes and WUE, and most in the years in which trees died.

Table 1: **Adding mortality improves the model skill for three of the four properties.** Coefficient of determination (R^2) of repeated leave-one-site-out predictions with climate and traits (C+T) and with mortality added (C+T+M); AutoGluon; Δ, change in R^2; 12- and 24-month predictor windows. ET_{max} does not improve.

	12 months			24 months		
EFP	C+T	C+T+M	Δ	C+T	C+T+M	Δ
GPP_{sat}	0.330	0.367	+0.037	0.317	0.370	+0.053
NEP_{max}	0.391	0.503	+0.112	0.394	0.502	+0.108
ET_{max}	0.423	0.425	+0.002	0.429	0.433	+0.004
WUE	0.252	0.312	+0.060	0.264	0.326	+0.062
Mean	0.349	0.402	+0.053	0.351	0.408	+0.057

Table 2: **Adding mortality reduces the prediction error (MAE), most in the years in which trees died.** Change in the MAE when the mortality predictors are added to climate and traits (C+T → C+T+M), AutoGluon, leave-one-site-out predictions: 100 (MAE_{C+T+M} / MAE_{C+T} − 1) %; negative = smaller error. Brackets: 95 % interval of a bootstrap that resamples whole sites (10 000 draws). Stars: one-sided paired permutation test with the two models swapped per site (10 000 draws), Benjamini–Hochberg corrected across all learners: $^{*}q < 0.05$, $^{**}q < 0.01$, $^{***}q < 0.001$. *Site-year level*: every site-year is one observation. *All years of the sites*: all sites, or only the sites whose maximum relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High), with all their years. *Only the years with mortality*: site-years whose own relative mortality exceeds 10.6 % or 24.0 %. Numbers under the column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5 maps.

		All years of the sites			Only the years with mortality	
		All sites (101 / 361)	Mid + High sites (60 / 225)	High sites (22 / 80)	> 10.6 % (60 / 160)	> 24.0 % (22 / 40)
AutoGluon 12 months	GPP_{sat}	−4.1 [−8.0, −0.2]	−6.1 [−10.9, −1.5]*	−7.2 [−18.1, +2.9]	−6.7 [−12.8, −0.7]*	−12.0 [−29.5, +4.9]
	NEP_{max}	−11.8 [−17.0, −6.3]**	−11.3 [−17.5, −5.1]**	−22.6 [−33.6, −11.1]**	−12.5 [−20.1, −4.9]*	−31.6 [−43.2, −16.0]*
	ET_{max}	−1.3 [−4.2, +1.3]	−2.1 [−5.3, +1.1]	−2.9 [−8.4, +2.1]	−3.9 [−7.7, +0.0]	−9.7 [−20.6, +2.7]
	WUE	−4.3 [−8.1, −0.2]	−6.8 [−11.3, −2.3]*	−10.9 [−17.8, −4.3]*	−7.2 [−12.2, −1.8]*	−16.9 [−25.2, −5.1]*
AutoGluon 24 months	GPP_{sat}	−5.7 [−9.5, −1.7]*	−8.3 [−12.6, −4.0]**	−7.1 [−17.2, +3.6]	−8.3 [−14.2, −2.6]*	−11.1 [−28.0, +8.1]
	NEP_{max}	−10.8 [−16.0, −5.1]**	−10.5 [−17.3, −3.4]*	−17.5 [−30.7, −2.1]	−11.6 [−19.8, −2.7]*	−29.2 [−42.9, −8.6]*
	ET_{max}	−2.0 [−5.0, +0.9]	−3.7 [−7.3, −0.1]	−5.3 [−13.1, +1.2]	−5.4 [−9.6, −1.3]*	−11.4 [−23.7, +3.2]
	WUE	−4.7 [−9.1, −0.1]	−6.4 [−11.5, −1.2]*	−14.8 [−23.9, −6.4]*	−8.2 [−14.5, −1.5]*	−21.8 [−34.5, −8.2]*

Without mortality, models over-predict

To find where these errors come from, we grouped the residuals (observed − predicted) by the relative mortality of each site-year (Fig. 2). Without mortality, the model predicted too high in high-mortality years. In site-years with more than 30 % relative mortality (21 site-years at 10 sites) it over-predicted NEP_{max} by 4.0 μmol m^{-2} s^{-1} (95 % interval 2.4–5.5), GPP_{sat} by 4.3 μmol m^{-2} s^{-1} (1.3–7.5) and WUE by 0.77 g C mm^{-1} (0.29–1.10), whereas the residuals were close to zero in years with little mortality. Adding mortality reduced these over-predictions by about two-thirds, to 1.3, 1.1 and 0.43, with intervals that included zero. This correction — the amount by which adding mortality lowered the prediction — was significant in the > 30 % class (GPP_{sat} +3.2, NEP_{max} +2.7 μmol m^{-2} s^{-1}, WUE +0.34 g C mm^{-1}; all $q < 0.01$, paired permutation test with whole sites swapped) and grew with relative mortality (mixed model with a random intercept per site, per +10 % relative mortality: +0.96, +0.75 and +0.10; all $q < 0.001$). In years with less than 10 % mortality the mortality predictors instead raised the predictions slightly. ET_{max} showed no such pattern. Across all 361 site-years, the residual of the model without mortality declined with relative mortality (per +10 %: NEP_{max} −0.77 μmol m^{-2} s^{-1}, WUE −0.18 g C mm^{-1}); for NEP_{max} and WUE this decline held when whole sites were resampled, when any single site was left out and when the three sites with more than 50 % mortality were removed (Fig. A21), whereas with mortality no trend remained. A model that ignores mortality thus systematically over-estimates carbon uptake and water-use efficiency where trees died.

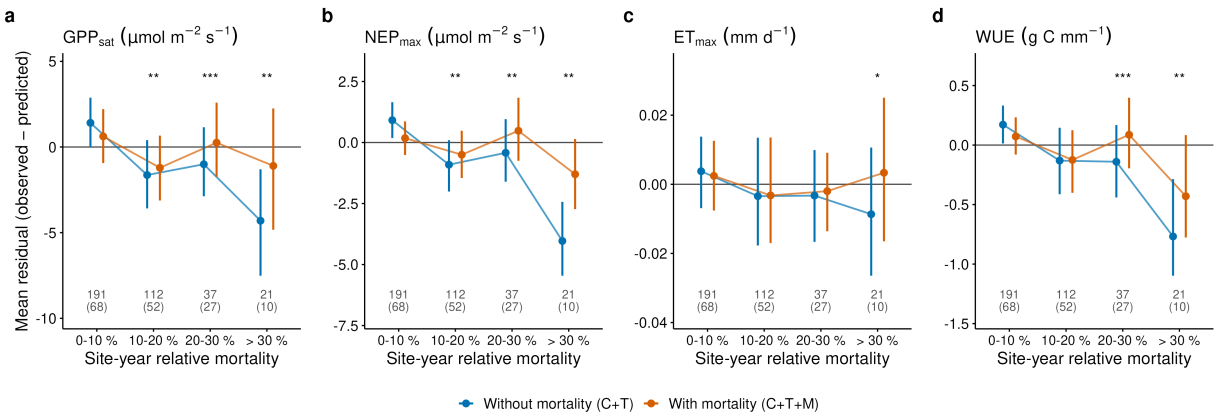


Fig. 2 | Without mortality, the model over-predicts the ecosystem functional properties in years with high mortality. Mean residual (observed − predicted) of the leave-one-site-out predictions of GPP_{sat} (**a**), NEP_{max} (**b**), ET_{max} (**c**) and WUE (**d**), grouped by the relative mortality of the site-year, for the model with climate and traits (C+T, blue) and with mortality added (C+T+M, vermillion); AutoGluon, 24-month window. Negative values mean that the model predicts too high. Points, mean; bars, 95 % bootstrap interval with whole sites resampled; numbers, site-years (sites) per class. Stars: adding mortality lowers the prediction in that class (one-sided paired permutation test with the two models swapped per site, Benjamini–Hochberg corrected; * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$).

Predictions drop where trees died

To see how the models use mortality, we summed the SHAP values of the mortality predictors for each site (Fig. 3). These contributions describe the fitted model, not a measured physiological response. At High sites the mortality predictors lowered the predicted GPP_{sat} at 86 %, NEP_{max} at 91 % and WUE at 86 % of the sites (medians −2.1 and −1.5 μmol m^{-2} s^{-1} for GPP_{sat} and NEP_{max}), whereas at Low sites they raised the predictions slightly at 90–98 % of the sites (medians +1.4 and +1.2 μmol m^{-2} s^{-1}); Mid sites were mixed (58–68 % negative). The contributions of High and Low+Mid sites differed for all four properties and both windows (Fig. 4; Mann–Whitney U test, all $q < 0.001$): at 24 months, GPP_{sat} −2.15 versus +0.66 μmol m^{-2} s^{-1}, NEP_{max} −1.51 versus +0.51 μmol m^{-2} s^{-1} and WUE −0.24 versus +0.07 g C mm^{-1} (medians). Because the sites are grouped by the variable whose contribution is measured, this comparison shows that the learned

relationship is consistent rather than providing independent evidence; its sign and size are the informative part. Together with Fig. 2, it shows that the models improve by lowering their predictions where trees died. Mortality accounted for 24 % (GPP_{sat}), 28 % (NEP_{max}), 16 % (ET_{max}) and 23 % (WUE) of the mean absolute SHAP, compared with 52–66 % for climate and 11–20 % for traits.

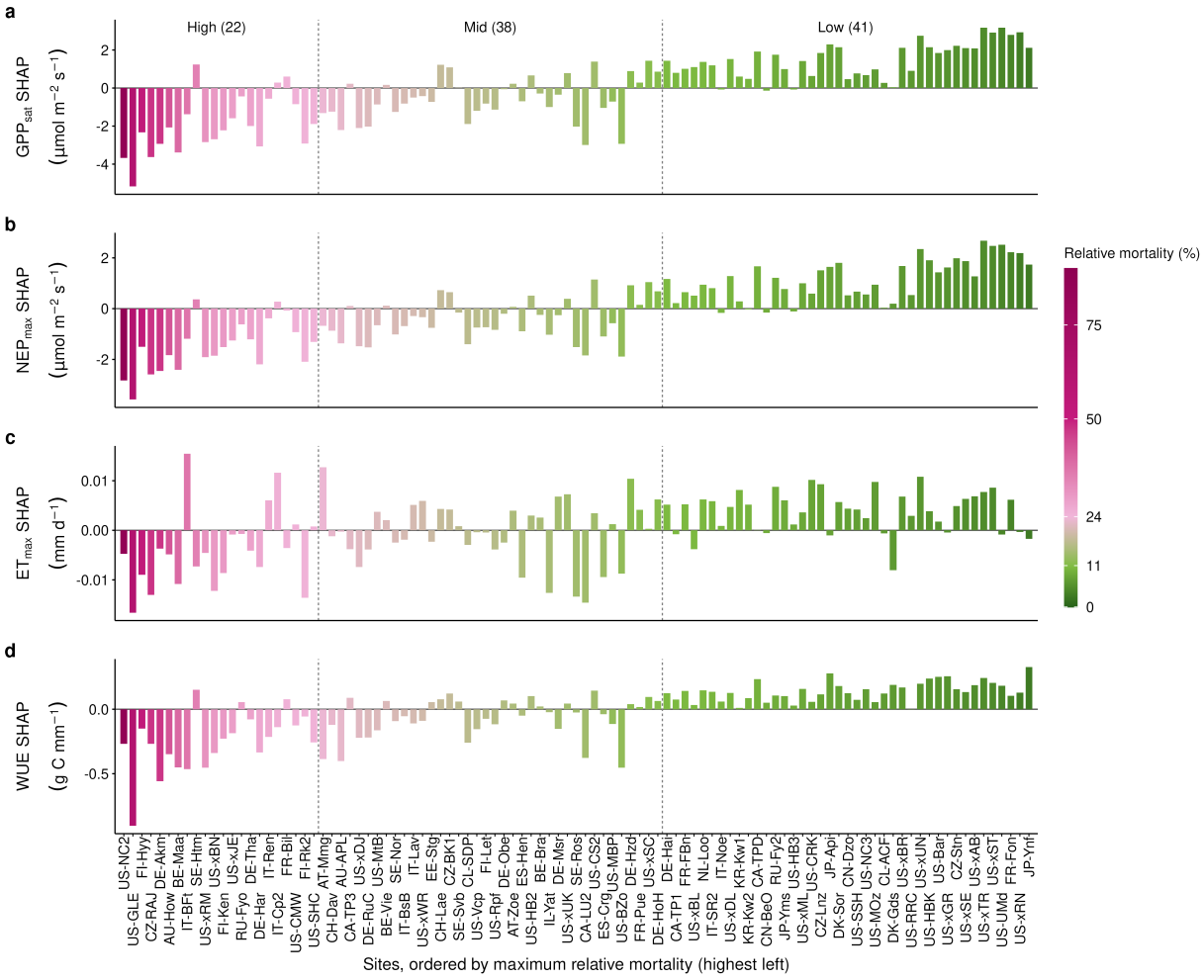


Fig. 3 | Mortality lowers the predicted carbon uptake and water-use efficiency at sites with high mortality and raises them slightly where mortality is low. Net contribution of the mortality predictors (sum of their SHAP values, mean over the years of each site) to the predicted GPP_{sat} (**a**), NEP_{max} (**b**), ET_{max} (**c**) and WUE (**d**), in the units of each property; model with climate, traits and mortality (C+T+M), AutoGluon, 24-month window. Negative values: mortality lowers the prediction at that site. The 101 sites are ordered by their maximum relative mortality (mean of five buffers of 100–500 m radius), highest on the left, and coloured from green (low) to pink (high), with the colour change anchored at the class breaks (dashed lines, 10.6 % and 24.0 %; number of sites per class in brackets). FLUXNET site codes below the panels.

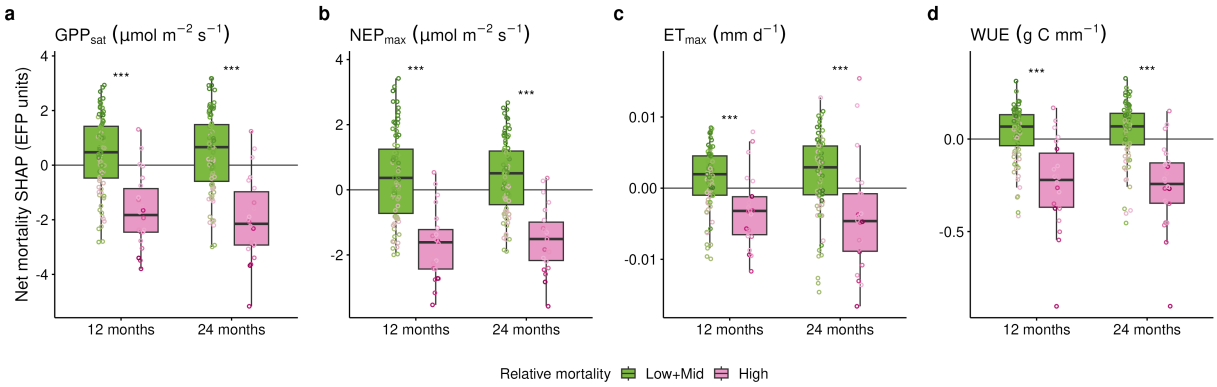


Fig. 4 | The direction of the mortality effect differs between high- and low-mortality sites. Net mortality SHAP per site (as in Fig. 3) at sites with high relative mortality (High, site maximum $> 24.0\,\%$, $n = 22$, pink) and at the other sites (Low+Mid, $n = 79$, green) for GPP_{sat} (**a**), NEP_{max} (**b**), ET_{max} (**c**) and WUE (**d**), 12- and 24-month windows; AutoGluon, C+T+M. Points are sites in their Fig. 3 colour; box colour, the Fig. 3 colour at the group's median relative mortality; boxes show the median and interquartile range, whiskers extend to 1.5 times the interquartile range. Stars: two-sided Mann–Whitney U test, Benjamini–Hochberg corrected across the eight tests ($^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$).

The effect scales with mortality

The contribution of mortality followed the amount of mortality closely (Fig. 5). Across sites, it declined with the site maximum of relative mortality for GPP_{sat} (Spearman $\rho = -0.81$), NEP_{max} ($\rho = -0.83$) and WUE ($\rho = -0.78$), and less for ET_{max} ($\rho = -0.47$; all $q < 0.001$); its sign changed at about 10–20 % relative mortality. The error reduction did not depend on tree cover ($\rho = 0.10$ to 0.18, not significant). The direction of the contribution was also related to tree cover ($\rho = 0.66$ to 0.75), but less closely than to relative mortality, and it was mostly negative in shrublands and wetlands and mostly positive in deciduous broadleaf, evergreen broadleaf and mixed forests, classes that contain few sites (Fig. A11). The effect of mortality on the predictions is therefore graded: the more trees died, the more the models lowered the predicted carbon uptake and WUE.

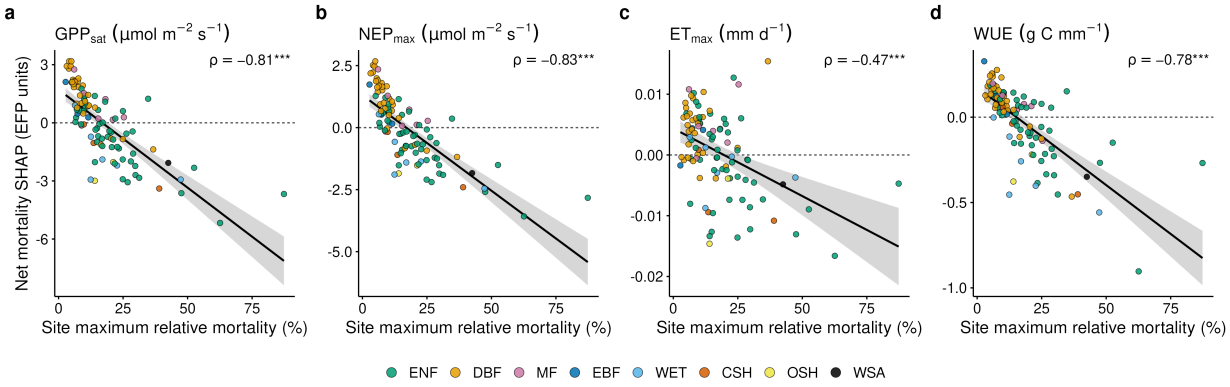


Fig. 5 | The direction of the mortality effect follows the amount of mortality. Net mortality SHAP per site (as in Fig. 3) against the site's maximum relative mortality for GPP_{sat} (**a**), NEP_{max} (**b**), ET_{max} (**c**) and WUE (**d**); AutoGluon, C+T+M, 24-month window; $n = 101$ sites coloured by IGBP class (abbreviations as in Fig. 1). Black line, linear fit with 95 % confidence band; dashed line, zero; ρ, Spearman rank correlation, Benjamini–Hochberg corrected across the four panels ($^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$).

The results are robust

XGBoost and LightGBM reached similar skill to AutoGluon (Table 3) and showed the same over-prediction without mortality in the $> 30\,\%$ class (NEP_{max} -3.6 to $-4.0\,\mu\mathrm{mol\,m^{-2}\,s^{-1}}$), with significant but about half as large corrections (Figs. A19 and A20). The results were the same with the

12-month window (Figs. A6 and A18), with site-level tests and with the root-mean-square error (Tables 4 and 6). Because some climate variables enter the calculation of the properties, we refitted each property without them (radiation for GPP_{sat}, air temperature for ET_{max} and WUE); R^2 changed by about 0.03 or less, and the error reductions, the over-prediction without mortality and its correction remained (Table 9 and Fig. A23). Finally, the five mortality metrics were strongly correlated: removing any single metric changed the error by at most about 1 %, and relative mortality alone gave most of the improvement (6.7–7.1 % of the 7.6–8.9 % NEP_{max} RMSE reduction of the full block with XGBoost and LightGBM; Figs. A14 and A15).

Discussion

Draft (about 1,000 words). Numbered citations; references in references.bib, each checked against the paper's abstract.

Satellite- and aerial-based maps of tree mortality made the functional properties of woody ecosystems more predictable, and the gain was concentrated where mortality occurred. Without mortality information, models over-predicted maximum net ecosystem productivity, light-saturated photosynthesis and water-use efficiency in years with high mortality, and this bias grew with the amount of mortality. Adding mortality reduced it by one- to two-thirds, depending on the learner, and the direction of the mortality effect followed the amount of mortality rather than vegetation type or tree cover. Mortality therefore carries information on carbon uptake that climate and plant traits do not.

The direction of the effect is consistent with what single-site studies have measured during mortality events. At a subalpine forest hit by spruce beetles, net ecosystem exchange diminished by about half once the attacked trees died, after an earlier phase in which hydraulic failure restricted gas exchange[1]. Across North American forests, net ecosystem production became a carbon loss after stand-replacing disturbances and returned to a sink within 10 to 20 years[2]. Our results generalize these site observations across 101 sites and several ecosystem types, and they suggest why mortality acts on productivity: the first and most important axis of ecosystem function, maximum productivity, is mostly explained by vegetation structure[3], and mortality changes that structure directly. Relative mortality, which combines standing dead trees with tree-cover loss, carried most of the information, and the direction of its effect was negative at most sites with high mortality and slightly positive at sites with little.

The effect was not uniform, and the literature shows why. At a lodgepole pine forest where beetles killed 30 to 78 % of the basal area, maximum CO_2 uptake did not change, because the remaining vegetation used light more efficiently and the changes in photosynthesis and respiration cancelled out[4]. In British Columbia, beetle-killed stands remained growing-season carbon sinks thanks to surviving trees and understory[5], and a synthesis of remote-sensing and flux data suggested that net ecosystem production may recover within about four years after climate-driven mortality[6]. Such compensation fits the shape of our results: errors changed little at low and moderate mortality and fell sharply only in the site-years with more than 30 % relative mortality, and the reduction over all site-years (11–12 % for maximum net ecosystem productivity) was much smaller than in high-mortality years (29–32 %). Our predictors included mortality of the current and the one or two preceding years, which allows models to capture the multi-year legacies that follow drought and mortality[7]; how quickly the functional properties recover after mortality remains to be tested with longer records.

Mortality improved water-use efficiency where it occurred but not maximum evapotranspiration. Measured responses of evapotranspiration to tree death differ between sites: evapotranspiration fell by 28–36 % during a spruce beetle outbreak[1], and evapotranspiration relative to precipitation fell by 30 % at a beetle-affected site[8], whereas reviews of die-off ecohydrology expect the loss of transpiration and interception to be partly offset by more evaporation from the soil and by surviving

and understory vegetation[9]. After drought-induced die-off in semi-arid basins, streamflow even decreased[10]. Maximum evapotranspiration, the 95th percentile of half-hourly evapotranspiration, is dominated by atmospheric demand and is therefore less sensitive to the loss of individual trees than carbon uptake. Mortality thus appears to act more on the carbon side of ecosystem function than on its water side.

These results matter for estimates of the forest carbon sink. Disturbances, many of them involving tree mortality, already shape the carbon balance at large scales: a mountain pine beetle outbreak turned the forests of British Columbia from a small carbon sink into a large source[11], increasing disturbances could reduce the carbon storage of European forests by about 500 Tg C in 2021–2030[12], drought and fire brought the Siberian forest carbon sink close to neutral[13], and stand-replacing disturbances account for a large part of global biomass turnover[14]. Data-driven upscaling of eddy-covariance fluxes does not include time since disturbance or site history among its predictors, which likely contributes to biased net ecosystem exchange patterns[15], and the representation of disturbance in vegetation models is a key reason why their estimates of the northern carbon sink disagree with atmospheric inversions[16]. Our models reproduce this blind spot: without mortality they over-estimated carbon uptake exactly where mortality was high. With tree mortality increasing worldwide[17–19] and forest disturbance regimes shifting towards younger and shorter stands[20,21], this bias is likely to grow unless mortality is represented explicitly.

Our study has limitations. The models are predictive, so they quantify how much mortality information improves predictions rather than the physiological mechanisms behind the change. Only 21 site-years at 10 sites had more than 30 % relative mortality; the strongest effects rest on these sites, although the over-prediction without mortality held when any single site or the three most extreme sites were left out. Mortality was derived from maps around each tower (100–500 m buffers), which do not match the varying flux footprint exactly, and errors in the maps add noise that probably weakens rather than strengthens the effects we report. Finally, the climate predictors cover only the period before the growing season, and leaving out the climate variables that enter the calculation of each property did not change the main results (Appendix A8), so the mortality effect does not arise from circularity with the climate data.

Knowing where and when trees die makes the carbon uptake of woody ecosystems more predictable. Without that knowledge, models systematically over-estimated carbon uptake in ecosystems with high mortality; with it, this bias shrank, and the correction scaled with the amount of mortality regardless of vegetation type or tree cover. Mortality maps now cover large areas at high resolution[22], and flux networks keep growing. Adding mortality to data-driven upscaling products and to the evaluation of land-surface models is a direct way to avoid over-estimating the carbon uptake of woody ecosystems with high mortality.

Methods

Eddy-covariance data. We used half-hourly FLUXNET data products processed with ONEFlux v1.3[23], downloaded for all available sites with the FLUXNET Shuttle on 12 May 2026. Net ecosystem exchange (NEE_VUT_REF) and gross primary productivity from night-time partitioning (GPP_NT_VUT_REF[24]) were used together with the tower meteorology (air temperature, vapour pressure deficit, incoming shortwave radiation, precipitation) and latent heat flux. Only measured or high-quality gap-filled half-hours (quality flags 0 and 1) entered the calculation of the ecosystem functional properties. The 101 sites of the final dataset come from AmeriFlux (44 sites), ICOS (29), the European Fluxes Database (18), JapanFlux[25] (3), KoFlux (2), ChinaFlux[26] (2), TERN-OzFlux[27,28] (2) and FLUXNET (1); the data of each site are cited individually (Data availability).

Ecosystem functional properties. For each site-year we calculated four ecosystem functional properties following Migliavacca et al.[3], within the growing season, defined as the days on which

daily gross primary productivity exceeded 30 % of its 95th percentile (bigleaf R package[29]). Light-saturated gross primary productivity (GPP_{sat}) was obtained by fitting a light-response curve of NEE against photosynthetic photon flux density (shortwave radiation × 2.11) in five-day windows of daytime data (shortwave radiation > 50 W m^{-2}) and taking the 95th percentile of the fitted saturation values per year[3,30]. Maximum net ecosystem productivity (NEP_{max}) is the 90th percentile of all half-hourly net ecosystem productivity (−NEE, day and night) in the growing season[3]. Maximum evapotranspiration (ET_{max}) is the 95th percentile of half-hourly evapotranspiration (mm d^{-1}), and water-use efficiency (WUE) the median ratio of gross primary productivity to evapotranspiration (g C mm^{-1})[29]; both used daytime half-hours with shortwave radiation > 200 W m^{-2} and friction velocity > 0.2 m s^{-1} and excluded half-hours within 24 h after precipitation of more than 0.1 mm, so that evaporation of intercepted water does not inflate evapotranspiration. Evapotranspiration was converted from latent heat flux with the air-temperature-dependent latent heat of vaporization[29].

Tree mortality. Tree mortality was derived from annual maps of standing deadwood cover and tree cover produced by deadtrees.earth from Sentinel-2 imagery, trained on centimetre-resolution aerial imagery with manually annotated dead trees[22] (map version 5; year 2016 excluded as unreliable). Within five nested circular buffers around each tower (radius 100, 200, 300, 400 and 500 m; pixels without a valid prediction masked) we calculated, per year: absolute deadwood (deadwood cover as a percentage of buffer area), relative deadwood (deadwood cover relative to tree cover, where tree cover ≥ 10 %, capped at 100 %), deadwood gain (year-on-year increase of deadwood cover, percentage points), relative tree-cover loss (percentage of the previous year's tree cover lost) and relative mortality, (deadwood + tree-cover loss) / (tree cover + tree-cover loss) × 100. Each metric entered the models for every buffer, for the year of the ecosystem functional property and for the one (12-month window) or two (24-month window) preceding years. For figures, sites were classified by their maximum relative mortality across years, averaged over the five buffers (Low < mean − 0.5 s.d., High > mean + 0.5 s.d. across sites).

Climate and plant traits. Climate predictors were calculated from the tower meteorology and cover only the period before the growing season: the 12 months ending in the month before the growing-season start (lag 1) and, for the 24-month window, the 12 months before that (lag 2). For every month we used the mean, 5th and 95th percentile of air temperature, vapour pressure deficit and shortwave radiation, and the precipitation sum. Plant traits were 23 community-weighted mean traits at 1 km resolution from global maps based on crowdsourced biodiversity observations and Earth observation[31], averaged within 500 m of each tower and weighted by area.

Site selection. Site-years were retained when all four ecosystem functional properties were available (2017–2025), lay in runs of at least two consecutive years and had complete climate and mortality predictors including their lags, which restricts the years to 2020–2025. Two plantation sites with extreme water-use efficiency (CN-Sb1, CN-Sb2) and site-years with negative water-use efficiency were removed. We kept sites with a mean tree cover of at least 30 % within 500 m. The final dataset has 101 sites and 361 site-years (median 4 years per site) of evergreen needleleaf (49 sites), deciduous broadleaf (30), mixed (7) and evergreen broadleaf (3) forests, wetlands (6), closed (3) and open shrublands (2) and woody savanna (1).

Models. For each property and window, two models were compared that differ only in the mortality predictors: climate and traits (C+T; 143 predictors for 12 months, 263 for 24 months) and climate, traits and mortality (C+T+M; 193 and 338 predictors). The main learner was AutoGluon-Tabular[32] (v1.6.3; default model zoo of gradient-boosted trees, random forests, extra trees and neural networks, five-fold bagging on site-disjoint folds, weighted ensemble without stacking). XGBoost[33] and LightGBM[34], each tuned with Optuna[35] (tree-structured Parzen estimator, 50 trials per property and window, five-fold cross-validation on site-disjoint groups), served as checks.

Cross-validation. Models were evaluated with repeated leave-one-site-out cross-validation: for every site, models were trained three times on a random 80 % of the remaining sites and the predictions for the held-out site were averaged. AutoGluon was refitted inside every fold, so that

model selection never saw the held-out site.

Model interpretation. Contributions of predictors were quantified with SHAP values[36]: exact TreeSHAP[37] for XGBoost and LightGBM and permutation-based SHAP for the AutoGluon ensemble. The net contribution of the mortality block is the sum of the signed SHAP values of all mortality predictors per site-year, averaged per site; negative values mean that mortality lowers the predicted property.

Statistics. The main test compares the mean absolute error (MAE) of C+T+M and C+T at the site-year level: every site-year is an observation, and the statistic is the difference of the pooled MAE (percentage change = ratio of the pooled errors). Because the years of one site are not independent, whole sites were resampled with replacement (10 000 bootstrap draws, 95 % percentile intervals) and, for a paired permutation test, the C+T and C+T+M predictions were swapped per site (10 000 draws, one-sided). The same tests were run for subsets of sites (High relative mortality) and of site-years (relative mortality of the year above the Mid or High break). At the site level, per-site errors were compared with a one-sided sign-flip permutation test on the log error ratios (100 000 flips). The relationship between the residuals (observed − predicted) and the total mortality of the site-year was tested with linear mixed models with a random intercept per site (nlme[38]), checked with a site-block bootstrap of the slope (1,000 draws), leave-one-site-out refits and without the three sites with more than 50 % relative mortality; the correction of the residual by mortality (residual of C+T+M minus residual of C+T) was tested per mortality class with the paired site-level permutation (100 000 draws). Differences in net mortality SHAP between High and Low+Mid sites were tested with Mann–Whitney U tests and associations with Spearman correlations. P values were corrected for multiple testing with the Benjamini–Hochberg procedure[39] within each table or figure. To test for circularity, each property was refitted without the climate variables that enter its own formula (shortwave radiation for GPP_{sat}, air temperature for ET_{max} and WUE).

Software. Analyses used R[40] (v4.3.3; xgboost, lightgbm, nlme, bigleaf) and Python (AutoGluon, Optuna, shap).

Use of large language models. **[DECISION: wording to be agreed by the authors; the journal requires any use of LLMs (e.g. for code or text assistance) to be documented here]**

Data availability

The FLUXNET data products used in this work were downloaded from the ICOS Carbon Portal and AmeriFlux via the FLUXNET Shuttle on 12 May 2026 and are cited per site[41–141]. Tree mortality maps are available from deadtrees.earth[22] and plant trait maps from Zenodo[31]. The derived site-year dataset and all model predictions are deposited at [repository, DOI].

Code availability

All code for dataset construction, model training (AutoGluon, XGBoost, LightGBM), statistics and figures is available at [repository, DOI].

Discussion references so far; Main and Methods still to be added (about 50 for an Article).

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Extended Data and Supplementary Information

The journal allows at most 10 Extended Data items (each cited in the main text); everything else goes to Supplementary Information. The A-sections below still need to be split into Extended Data and Supplementary.

A1. Learner comparison

- Same data, predictors, windows and cross-validation for all three learners.
- AutoGluon best in 7 of 16 tasks; mean R^2 (C+T+M) 0.402 / 0.408 (12 / 24 m) vs XGBoost 0.385 / 0.406 and LightGBM 0.388 / 0.400.
- The effect of mortality is very similar: NEP_{max} improves for every learner; overall RMSE change for GPP_{sat} −2.8 to −4.0 % for all learners; AutoGluon gives a somewhat larger WUE effect (−4 % vs −1 to −4 %).

Table 3: **The three learners reach similar skill.** R^2 of repeated leave-one-site-out predictions of the C+T+M model for AutoGluon, XGBoost and LightGBM (Optuna-tuned); bold, best learner per property and window; bottom rows, mean over the four properties with (C+T+M) and without mortality (C+T).

	12 months			24 months		
EFP	AutoGluon	XGBoost	LightGBM	AutoGluon	XGBoost	LightGBM
GPP_{sat}	**0.367**	0.334	0.346	0.370	**0.387**	0.347
NEP_{max}	**0.503**	0.473	0.492	0.502	0.514	**0.540**
ET_{max}	0.425	0.439	**0.443**	0.433	0.447	**0.453**
WUE	**0.312**	0.292	0.270	**0.326**	0.277	0.261
Mean, C+T+M	**0.402**	0.385	0.388	**0.408**	0.406	0.400
Mean, C+T	**0.349**	0.337	0.347	0.351	**0.365**	0.356

A2. Per-site error: figures and site-level tests

- Per-site MAE without and with mortality; % = geometric-mean change of the per-site MAE, stars = one-sided paired sign-flip permutation test on the per-site log MAE ratios (100 000 flips, BH-FDR across the figure). The log ratio stops sites with an almost perfect C+T prediction from dominating (one-year sites, e.g. KR-Kw1).
- NEP_{max} MAE −12 / −8 % across all sites and −30 / −17 % at High sites (12 / 24 months); WUE −11 / −15 % at High sites.



Fig. A1 | Per-site error without and with mortality (AutoGluon). MAE of every site for C+T (cyan) and C+T+M (pink), all 101 sites. Percentages, geometric-mean change of the per-site MAE; stars, one-sided paired sign-flip permutation test on the per-site log MAE ratios, Benjamini–Hochberg corrected.

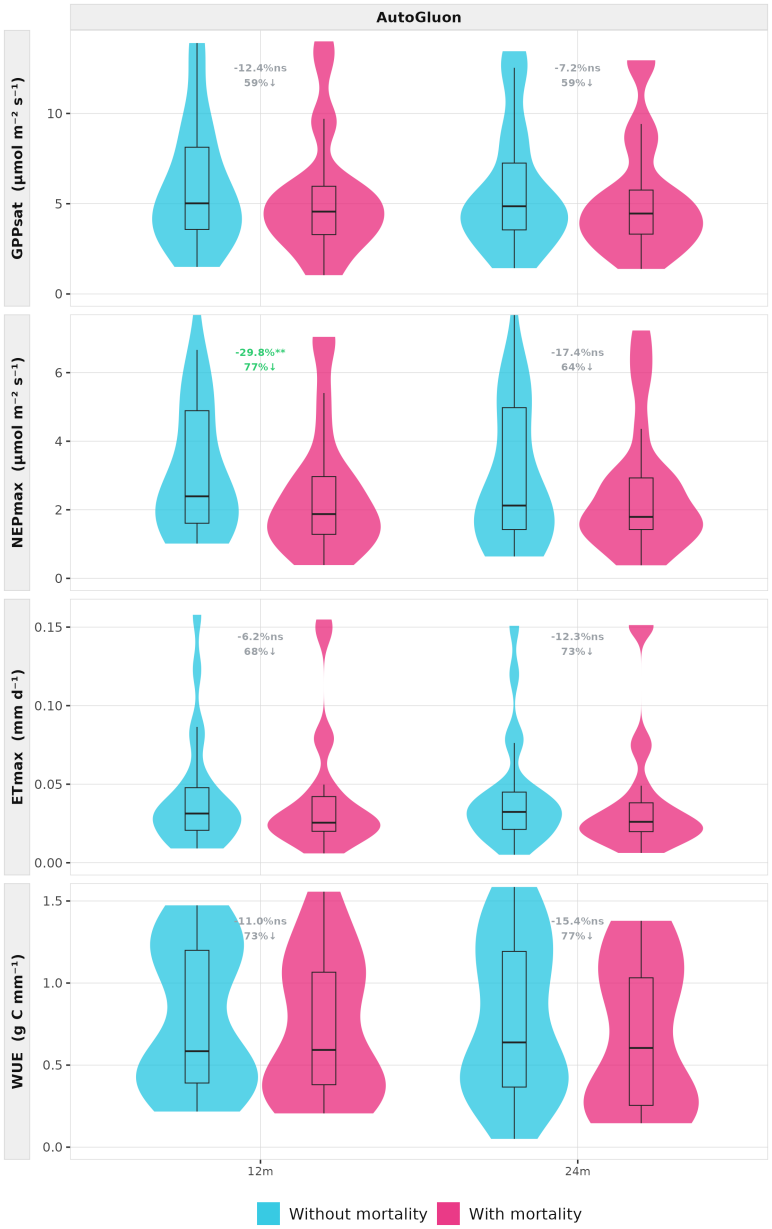


Fig. A2 | As Fig. A1 for the 22 sites with high relative mortality (models trained on all sites).

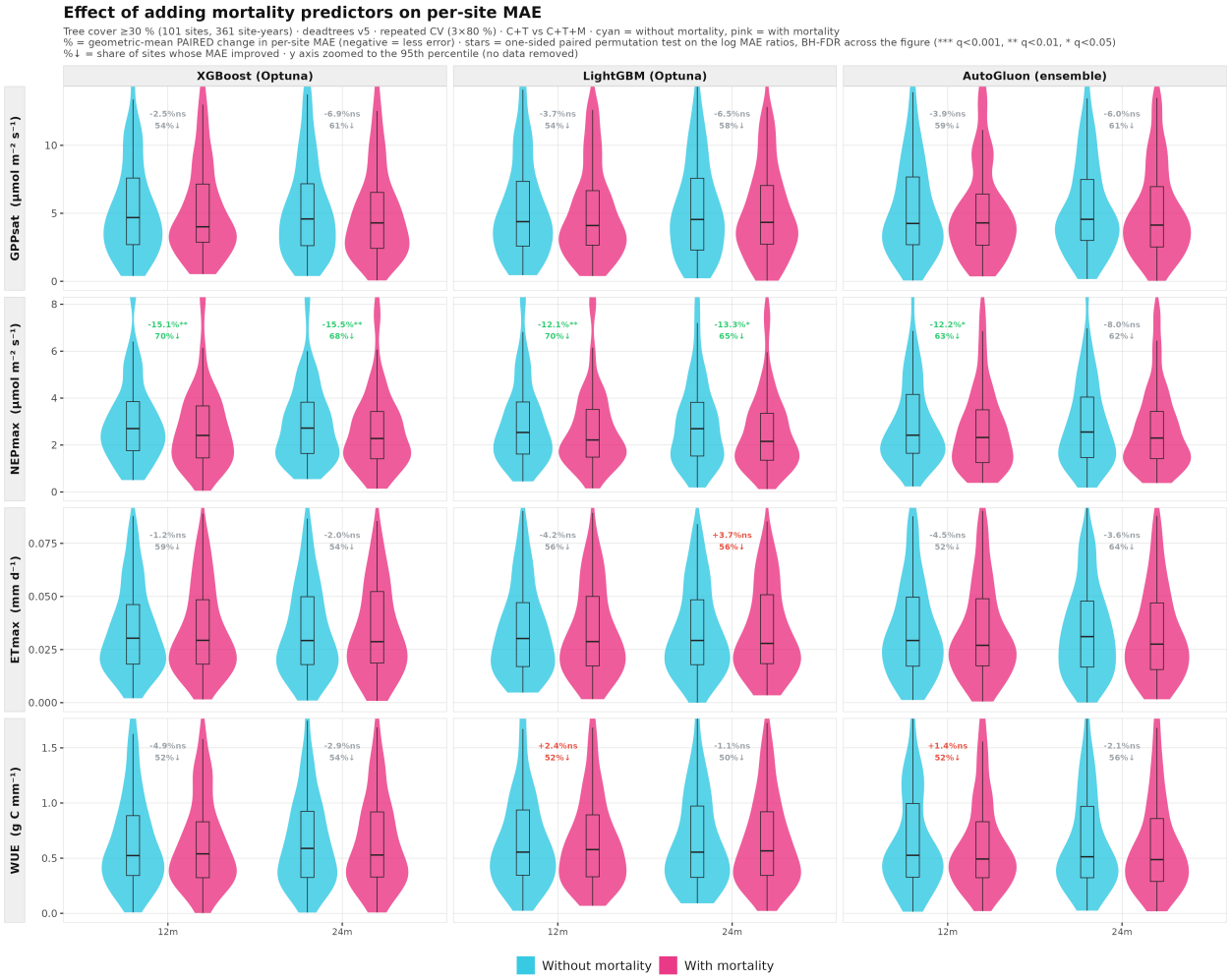


Fig. A3 | As Fig. A1 for the three learners (AutoGluon, XGBoost, LightGBM).

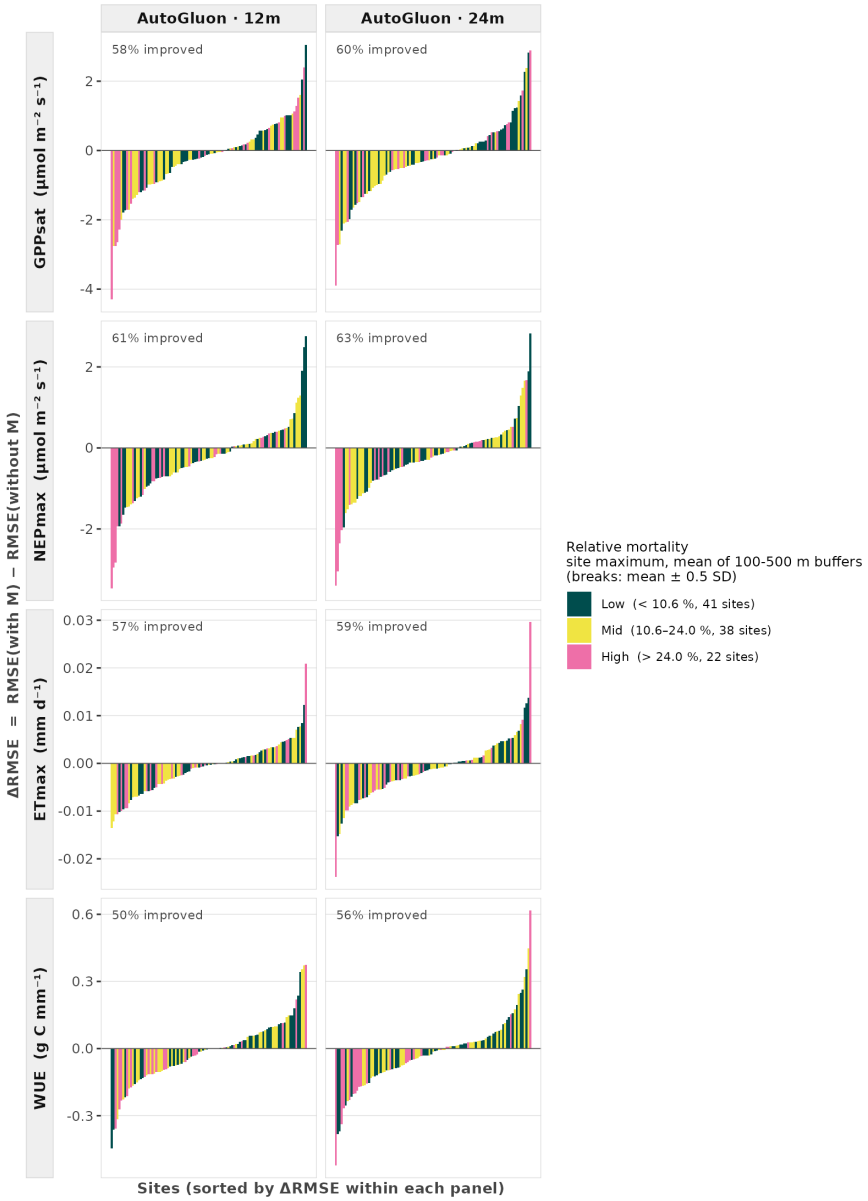


Fig. A4 | Per-site change in RMSE when mortality is added (AutoGluon). Negative values, smaller error; bars coloured by relative-mortality class.

Table 4: **Adding mortality reduces the per-site prediction error** (site level; per-site errors in Fig. A1). For every site, MAE of the model with climate, traits and mortality (C+T+M) is compared with the model without mortality (C+T), both evaluated by repeated leave-one-site-out cross-validation (3 × 80%). ΔMAE = geometric mean over sites of the paired error ratio, $(\exp(\text{mean}_{\text{sites}} \ln(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}})) - 1) \times 100$; negative values mean that mortality reduces the error. Improved = share of sites whose MAE decreased. Sites are grouped by their maximum relative mortality across years (averaged over the five buffers, 100–500 m); High > mean + 0.5 SD (24.0%), Low < mean − 0.5 SD (10.6%), Mid in between. Stars: one-sided paired permutation test on the per-site log error ratios (100 000 random sign flips; C+T+M < C+T), Benjamini–Hochberg corrected across all 72 tests of the three learners (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$). Tree cover ≥ 30%, 101 sites, 361 site-years, deadtrees v5.

Model	EFP		All sites (n = 101)	Mid + High (n = 60)	High (n = 22)
AutoGluon 12 months	GPP_{sat}	ΔMAE (%)	−3.9	−8.6*	−12.4
		Improved (%)	59	62	59
	NEP_{max}	ΔMAE (%)	−12.2*	−14.5**	−29.8**
		Improved (%)	63	62	77
	ET_{max}	ΔMAE (%)	−4.5	−5.6	−6.2
		Improved (%)	52	58	68
	WUE	ΔMAE (%)	+1.4	−5.2	−11.0
		Improved (%)	52	60	73
AutoGluon 24 months	GPP_{sat}	ΔMAE (%)	−6.0	−9.0**	−7.2
		Improved (%)	61	72	59
	NEP_{max}	ΔMAE (%)	−8.0	−9.1	−17.4
		Improved (%)	62	57	64
	ET_{max}	ΔMAE (%)	−3.6	−6.9*	−12.3
		Improved (%)	64	72	73
	WUE	ΔMAE (%)	−2.1	−6.1	−15.4
		Improved (%)	56	63	77

Table 5: **Adding mortality reduces the prediction error (MAE), most in the years in which trees died.** Change in the MAE when the mortality predictors are added to climate and traits (C+T → C+T+M), AutoGluon, leave-one-site-out predictions: $100\,(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}} - 1)$ %; negative = smaller error. Brackets: 95 % interval of a bootstrap that resamples whole sites (10 000 draws). Stars: one-sided paired permutation test with the two models swapped per site (10 000 draws), Benjamini–Hochberg corrected across all learners: * $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$. *Site level*: one error per site (mean over its years). *Site-year level*: every site-year is one observation. *All years of the sites*: all sites, or only the sites whose maximum relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High), with all their years. *Only the years with mortality*: site-years whose own relative mortality exceeds 10.6 % or 24.0 %. Numbers under the column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5 maps.

		Site level	Site-year level				
			All years of the sites			Only the years with mortality	
		All sites (101 sites)	All sites (101 / 361)	Mid + High sites (60 / 225)	High sites (22 / 80)	> 10.6 % (60 / 160)	> 24.0 % (22 / 40)
AutoGluon 12 months	GPP_{sat}	−4.6 [−8.6, −0.5]*	−4.1 [−8.0, −0.2]	−6.1 [−10.9, −1.5]*	−7.2 [−18.1, +2.9]	−6.7 [−12.8, −0.7]*	−12.0 [−29.5, +4.9]
	NEP_{max}	−10.1 [−15.5, −4.4]**	−11.8 [−17.0, −6.3]**	−11.3 [−17.5, −5.1]**	−22.6 [−33.6, −11.1]**	−12.5 [−20.1, −4.9]*	−31.6 [−43.2, −16.0]*
	ET_{max}	−2.5 [−5.6, +0.5]	−1.3 [−4.2, +1.3]	−2.1 [−5.3, +1.1]	−2.9 [−8.4, +2.1]	−3.9 [−7.7, +0.0]	−9.7 [−20.6, +2.7]
	WUE	−3.2 [−7.3, +1.3]	−4.3 [−8.1, −0.2]	−6.8 [−11.3, −2.3]*	−10.9 [−17.8, −4.3]*	−7.2 [−12.2, −1.8]*	−16.9 [−25.2, −5.1]*
AutoGluon 24 months	GPP_{sat}	−5.5 [−9.4, −1.4]*	−5.7 [−9.5, −1.7]*	−8.3 [−12.6, −4.0]**	−7.1 [−17.2, +3.6]	−8.3 [−14.2, −2.6]*	−11.1 [−28.0, +8.1]
	NEP_{max}	−9.3 [−14.4, −3.7]**	−10.8 [−16.0, −5.1]**	−10.5 [−17.3, −3.4]*	−17.5 [−30.7, −2.1]	−11.6 [−19.8, −2.7]*	−29.2 [−42.9, −8.6]*
	ET_{max}	−3.8 [−7.3, −0.1]*	−2.0 [−5.0, +0.9]	−3.7 [−7.3, −0.1]	−5.3 [−13.1, +1.2]	−5.4 [−9.6, −1.3]*	−11.4 [−23.7, +3.2]
	WUE	−3.6 [−8.3, +1.6]	−4.7 [−9.1, −0.1]	−6.4 [−11.5, −1.2]*	−14.8 [−23.9, −6.4]*	−8.2 [−14.5, −1.5]*	−21.8 [−34.5, −8.2]*

Table 6: **Adding mortality reduces the prediction error (RMSE), most in the years in which trees died.** Change in the RMSE when the mortality predictors are added to climate and traits (C+T → C+T+M), AutoGluon, leave-one-site-out predictions: $100\,(\mathrm{RMSE}_{\mathrm{C+T+M}} / \mathrm{RMSE}_{\mathrm{C+T}} - 1)\,\%$; negative = smaller error. Brackets: 95 % interval of a bootstrap that resamples whole sites (10 000 draws). Stars: one-sided paired permutation test with the two models swapped per site (10 000 draws), Benjamini–Hochberg corrected across all learners: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$. *Site level*: one error per site (mean over its years). *Site-year level*: every site-year is one observation. *All years of the sites*: all sites, or only the sites whose maximum relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High), with all their years. *Only the years with mortality*: site-years whose own relative mortality exceeds 10.6 % or 24.0 %. Numbers under the column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5 maps.

		Site level	Site-year level				
			All years of the sites			Only the years with mortality	
		All sites (101 sites)	All sites (101 / 361)	Mid + High sites (60 / 225)	High sites (22 / 80)	> 10.6 % (60 / 160)	> 24.0 % (22 / 40)
AutoGluon 12 months	$\mathrm{GPP}_{\mathrm{sat}}$	−3.6 [−7.2, +0.1]	−2.8 [−6.1, +0.5]	−4.2 [−8.5, −0.1]	−4.4 [−15.2, +4.5]	−4.4 [−10.0, +0.9]	−10.1 [−27.6, +5.8]
	$\mathrm{NEP}_{\mathrm{max}}$	−9.3 [−14.4, −3.9]**	−9.7 [−14.6, −4.7]**	−10.3 [−16.4, −4.3]*	−21.0 [−33.7, −7.7]*	−11.8 [−19.7, −4.3]*	−32.6 [−45.1, −16.0]*
	$\mathrm{ET}_{\mathrm{max}}$	−2.4 [−5.1, +0.3]	−0.2 [−2.6, +2.1]	−0.4 [−3.4, +2.3]	+0.0 [−5.9, +5.6]	−0.7 [−4.7, +3.5]	−0.7 [−17.2, +17.2]
	WUE	−2.2 [−6.0, +1.8]	−4.1 [−7.3, −1.2]*	−4.9 [−9.1, −1.6]*	−7.5 [−14.3, −1.5]*	−5.7 [−10.3, −1.4]*	−14.3 [−22.3, −2.7]
AutoGluon 24 months	$\mathrm{GPP}_{\mathrm{sat}}$	−4.3 [−7.9, −0.6]*	−4.0 [−7.3, −0.2]*	−6.0 [−9.8, −2.5]*	−4.2 [−14.0, +4.7]	−6.0 [−11.2, −1.2]*	−9.8 [−25.4, +7.0]
	$\mathrm{NEP}_{\mathrm{max}}$	−8.3 [−13.3, −2.9]**	−9.4 [−14.2, −4.2]*	−10.6 [−16.7, −4.3]*	−19.5 [−32.3, −5.7]*	−11.8 [−19.5, −3.8]*	−31.0 [−44.0, −13.1]*
	$\mathrm{ET}_{\mathrm{max}}$	−3.1 [−6.3, +0.2]	−0.4 [−3.1, +2.6]	−1.4 [−5.1, +2.2]	+0.6 [−7.8, +7.9]	−1.6 [−6.4, +2.9]	+0.4 [−19.3, +22.0]
	WUE	−3.0 [−7.2, +1.6]	−4.3 [−7.4, −1.2]*	−4.9 [−8.8, −1.2]*	−9.2 [−16.3, −2.4]*	−6.7 [−11.4, −1.5]*	−16.9 [−24.6, −6.5]*

Table 7: **Adding mortality reduces the prediction error (MAE), most in the years in which trees died.** Change in the MAE when the mortality predictors are added to climate and traits (C+T → C+T+M), XGBoost and LightGBM, leave-one-site-out predictions: $100\,(\mathrm{MAE}_{\mathrm{C+T+M}} / \mathrm{MAE}_{\mathrm{C+T}} - 1)\,\%$; negative = smaller error. Brackets: 95 % interval of a bootstrap that resamples whole sites (10 000 draws). Stars: one-sided paired permutation test with the two models swapped per site (10 000 draws), Benjamini–Hochberg corrected across all learners: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$. *Site level*: one error per site (mean over its years). *Site-year level*: every site-year is one observation. *All years of the sites*: all sites, or only the sites whose maximum relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High), with all their years. *Only the years with mortality*: site-years whose own relative mortality exceeds 10.6 % or 24.0 %. Numbers under the column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5 maps.

		Site level	Site-year level				
			All years of the sites			Only the years with mortality	
		All sites (101 sites)	All sites (101 / 361)	Mid + High sites (60 / 225)	High sites (22 / 80)	> 10.6 % (60 / 160)	> 24.0 % (22 / 40)
XGBoost 12 months	$\mathrm{GPP}_{\mathrm{sat}}$	−3.5 [−6.6, −0.3]*	−3.2 [−6.3, +0.1]	−2.6 [−6.1, +1.1]	−5.3 [−12.7, +2.5]	−3.7 [−8.1, +0.9]	−10.6 [−18.1, +0.7]
	$\mathrm{NEP}_{\mathrm{max}}$	−10.3 [−15.1, −5.0]**	−10.7 [−15.7, −4.9]**	−10.4 [−16.0, −4.7]**	−17.4 [−27.7, −7.7]**	−12.3 [−18.8, −5.8]**	−28.8 [−37.3, −17.6]**
	$\mathrm{ET}_{\mathrm{max}}$	−1.4 [−3.5, +0.7]	−0.9 [−3.0, +1.3]	−1.2 [−3.6, +1.3]	+0.1 [−2.9, +3.8]	−2.4 [−5.0, +0.2]	−1.9 [−6.1, +2.7]
	WUE	−2.8 [−6.9, +1.7]	−2.9 [−6.3, +0.7]	−3.7 [−7.6, +0.6]	−6.9 [−13.6, +0.5]	−4.2 [−8.8, +0.9]	−12.7 [−19.2, −5.9]*
XGBoost 24 months	$\mathrm{GPP}_{\mathrm{sat}}$	−5.8 [−9.9, −1.6]*	−5.4 [−9.8, −1.0]*	−6.4 [−11.3, −1.7]*	−6.6 [−16.2, +2.9]	−7.8 [−13.4, −2.4]*	−6.4 [−18.2, +5.1]
	$\mathrm{NEP}_{\mathrm{max}}$	−10.1 [−14.6, −5.6]**	−9.5 [−14.0, −4.8]**	−7.5 [−12.8, −2.3]*	−17.2 [−27.9, −8.3]**	−9.7 [−15.5, −4.2]**	−19.8 [−30.6, −10.7]**
	$\mathrm{ET}_{\mathrm{max}}$	−0.3 [−2.8, +2.2]	−0.4 [−3.0, +2.1]	−1.8 [−4.7, +1.2]	−2.0 [−5.9, +1.7]	−2.5 [−5.6, +0.7]	−4.0 [−8.2, +0.6]
	WUE	−1.8 [−6.3, +3.3]	−2.3 [−6.1, +2.1]	−2.7 [−7.0, +2.5]	−5.3 [−12.8, +4.6]	−3.1 [−8.3, +3.5]	−11.7 [−19.7, −3.1]*
LightGBM 12 months	$\mathrm{GPP}_{\mathrm{sat}}$	−4.0 [−7.1, −0.8]*	−3.3 [−6.4, +0.1]	−3.4 [−7.1, +0.2]	−6.6 [−14.0, +0.4]	−5.8 [−10.0, −1.6]*	−10.3 [−17.7, −3.1]*
	$\mathrm{NEP}_{\mathrm{max}}$	−9.5 [−14.8, −3.8]**	−9.6 [−15.0, −3.4]**	−8.8 [−14.8, −2.4]*	−14.7 [−25.4, −4.1]*	−10.7 [−17.5, −3.1]*	−24.7 [−35.2, −11.3]*
	$\mathrm{ET}_{\mathrm{max}}$	−2.1 [−4.4, +0.3]	−1.4 [−3.7, +1.0]	−2.2 [−4.9, +0.7]	−0.2 [−4.0, +4.6]	−3.4 [−6.4, −0.4]*	−3.8 [−9.5, +2.4]
	WUE	+0.0 [−3.7, +4.2]	−1.6 [−4.7, +2.2]	−2.9 [−6.7, +1.7]	−2.8 [−10.0, +6.5]	−2.8 [−7.6, +3.1]	−7.9 [−15.0, +0.4]
LightGBM 24 months	$\mathrm{GPP}_{\mathrm{sat}}$	−3.8 [−6.7, −0.8]*	−3.1 [−6.3, +0.1]	−3.9 [−7.5, −0.3]	−5.2 [−12.0, +1.8]	−4.3 [−8.6, +0.1]	−5.6 [−15.4, +5.9]
	$\mathrm{NEP}_{\mathrm{max}}$	−12.2 [−17.6, −6.6]**	−11.4 [−16.6, −5.7]**	−9.9 [−15.9, −4.0]*	−14.0 [−24.9, −4.0]*	−10.3 [−17.3, −3.3]*	−16.0 [−28.5, −3.1]*
	$\mathrm{ET}_{\mathrm{max}}$	−1.6 [−3.7, +0.6]	−1.3 [−3.4, +1.0]	−1.7 [−4.1, +0.9]	−0.6 [−3.7, +3.3]	−2.3 [−5.0, +0.6]	−3.8 [−8.5, +2.0]
	WUE	−0.6 [−4.9, +4.0]	−1.1 [−4.7, +2.9]	−0.8 [−4.6, +3.5]	−3.8 [−10.5, +4.4]	−2.0 [−6.9, +3.5]	−7.6 [−15.5, +0.3]

Table 8: **Adding mortality reduces the per-site prediction error** (site level; per-site errors in Fig. A1). For every site, MAE of the model with climate, traits and mortality (C+T+M) is compared with the model without mortality (C+T), both evaluated by repeated leave-one-site-out cross-validation (3 × 80%). ΔMAE = geometric mean over sites of the paired error ratio, $(\exp(\text{mean}_{\text{sites}} \ln(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}})) - 1) \times 100$; negative values mean that mortality reduces the error. Improved = share of sites whose MAE decreased. Sites are grouped by their maximum relative mortality across years (averaged over the five buffers, 100–500 m); High > mean + 0.5 SD (24.0%), Low < mean − 0.5 SD (10.6%), Mid in between. Stars: one-sided paired permutation test on the per-site log error ratios (100 000 random sign flips; C+T+M < C+T), Benjamini–Hochberg corrected across all 72 tests of the three learners (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$). Tree cover ≥ 30%, 101 sites, 361 site-years, deadtrees v5.

Model	EFP		All sites (n = 101)	Mid + High (n = 60)	High (n = 22)
XGBoost 12 months	GPP_{sat}	ΔMAE (%)	−2.5	+1.4	+0.2
		Improved (%)	54	57	59
	NEP_{max}	ΔMAE (%)	−15.1**	−13.0**	−18.4*
		Improved (%)	70	68	73
	ET_{max}	ΔMAE (%)	−1.2	−3.9	−0.5
		Improved (%)	59	63	64
	WUE	ΔMAE (%)	−4.9	−6.8	−25.2*
		Improved (%)	52	53	68
XGBoost 24 months	GPP_{sat}	ΔMAE (%)	−6.9	−12.7*	−18.7
		Improved (%)	61	65	68
	NEP_{max}	ΔMAE (%)	−15.5**	−16.5**	−29.4**
		Improved (%)	68	63	73
	ET_{max}	ΔMAE (%)	−2.0	−3.1	−5.0
		Improved (%)	54	63	68
	WUE	ΔMAE (%)	−2.9	−3.9	−20.9
		Improved (%)	54	57	68
LightGBM 12 months	GPP_{sat}	ΔMAE (%)	−3.7	−2.6	−8.5
		Improved (%)	54	58	68
	NEP_{max}	ΔMAE (%)	−12.1**	−10.6*	−17.9*
		Improved (%)	70	70	77
	ET_{max}	ΔMAE (%)	−4.2	−5.8	−1.3
		Improved (%)	56	62	50
	WUE	ΔMAE (%)	+2.4	−2.0	−9.6
		Improved (%)	52	62	55
LightGBM 24 months	GPP_{sat}	ΔMAE (%)	−6.5	−7.1*	−10.6
		Improved (%)	58	62	64
	NEP_{max}	ΔMAE (%)	−13.3*	−16.6**	−22.1*
		Improved (%)	65	63	68
	ET_{max}	ΔMAE (%)	+3.7	−1.2	−1.1
		Improved (%)	56	58	59
	WUE	ΔMAE (%)	−1.1	+0.2	−8.4
		Improved (%)	50	48	64

A3. Direction per site: site names, 12 months and XGBoost / LightGBM

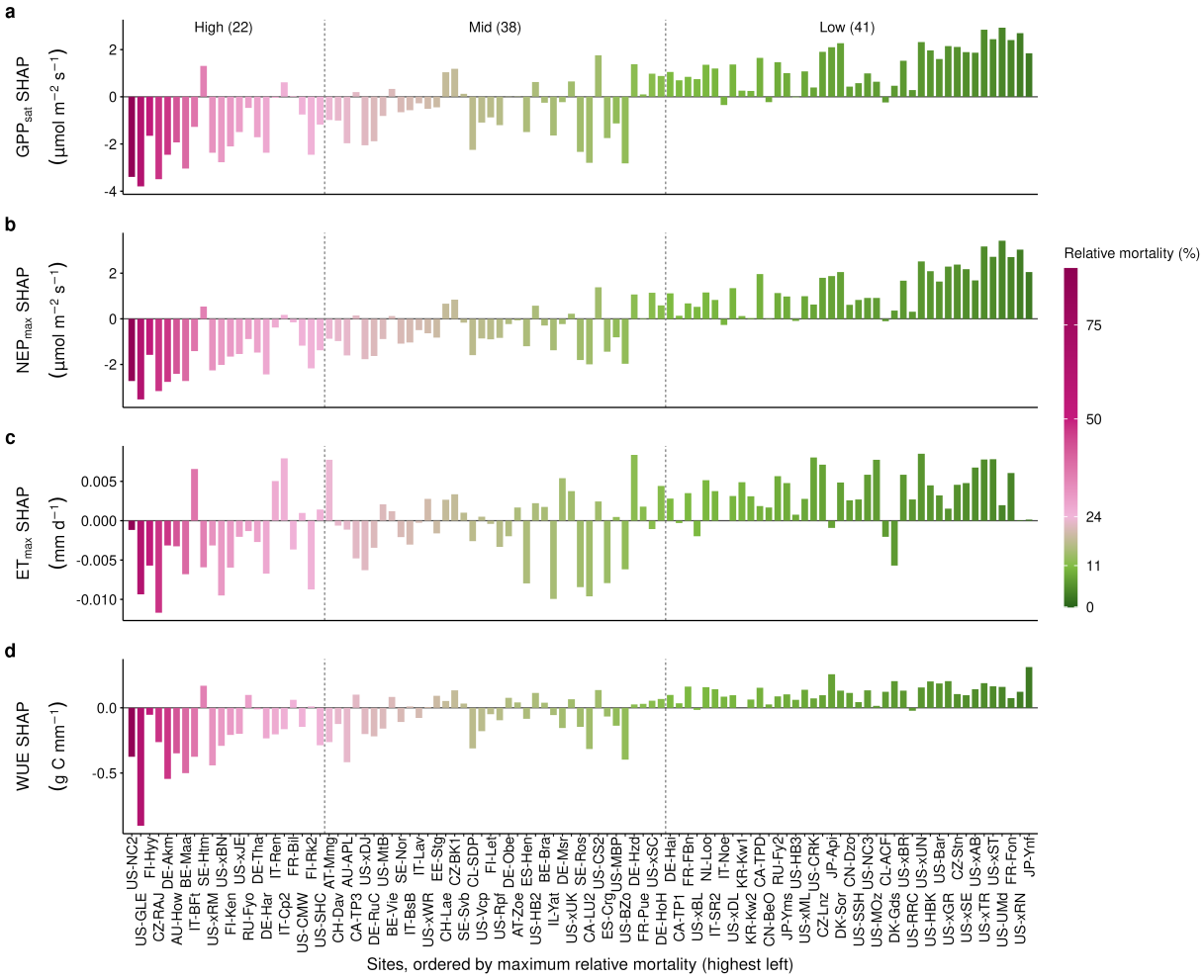


Fig. A6 | As Fig. 3 for the 12-month window (AutoGluon).

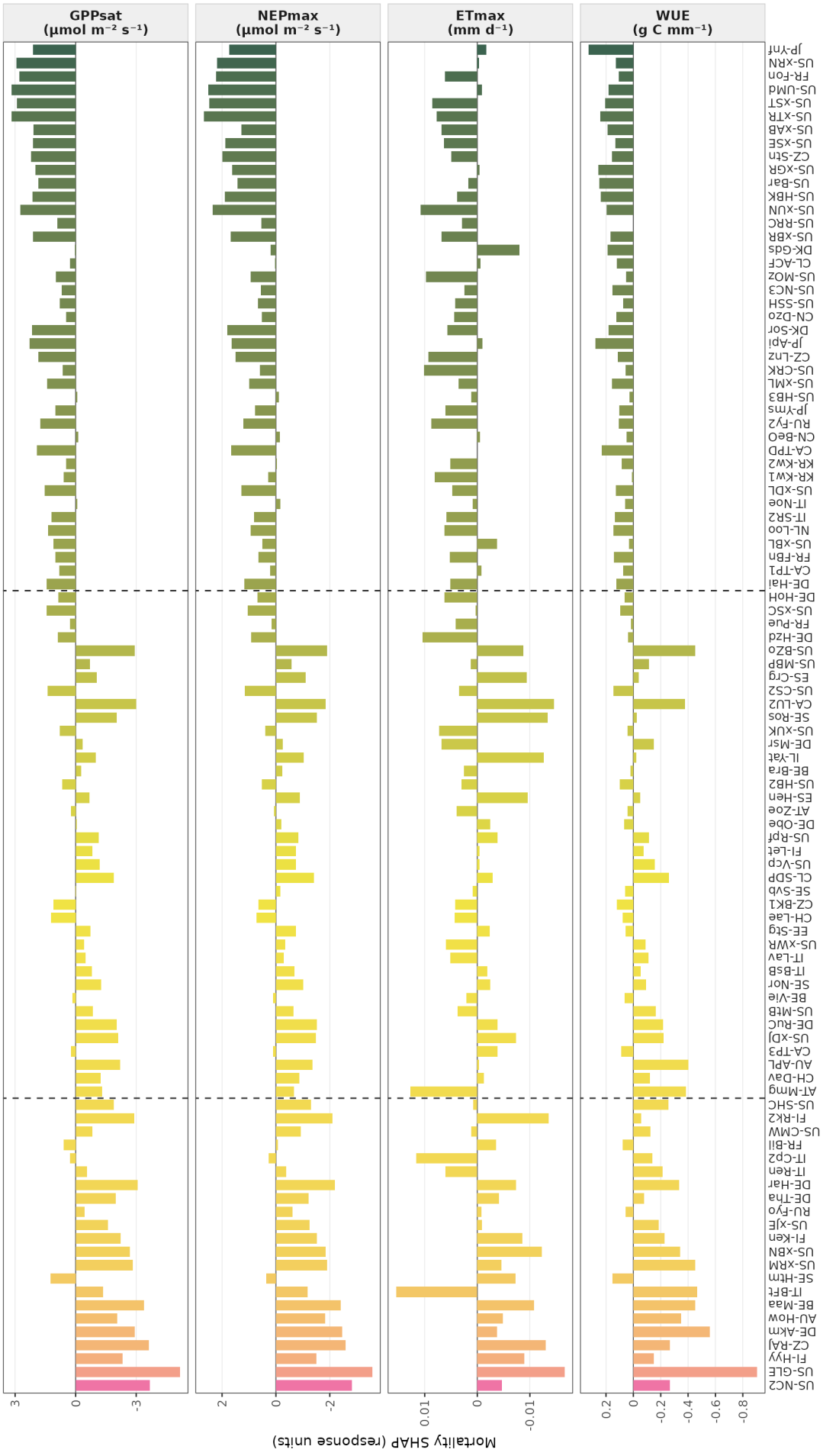


Fig. A5 | As Fig. 3 in landscape layout with site names (AutoGluon, 24 months).

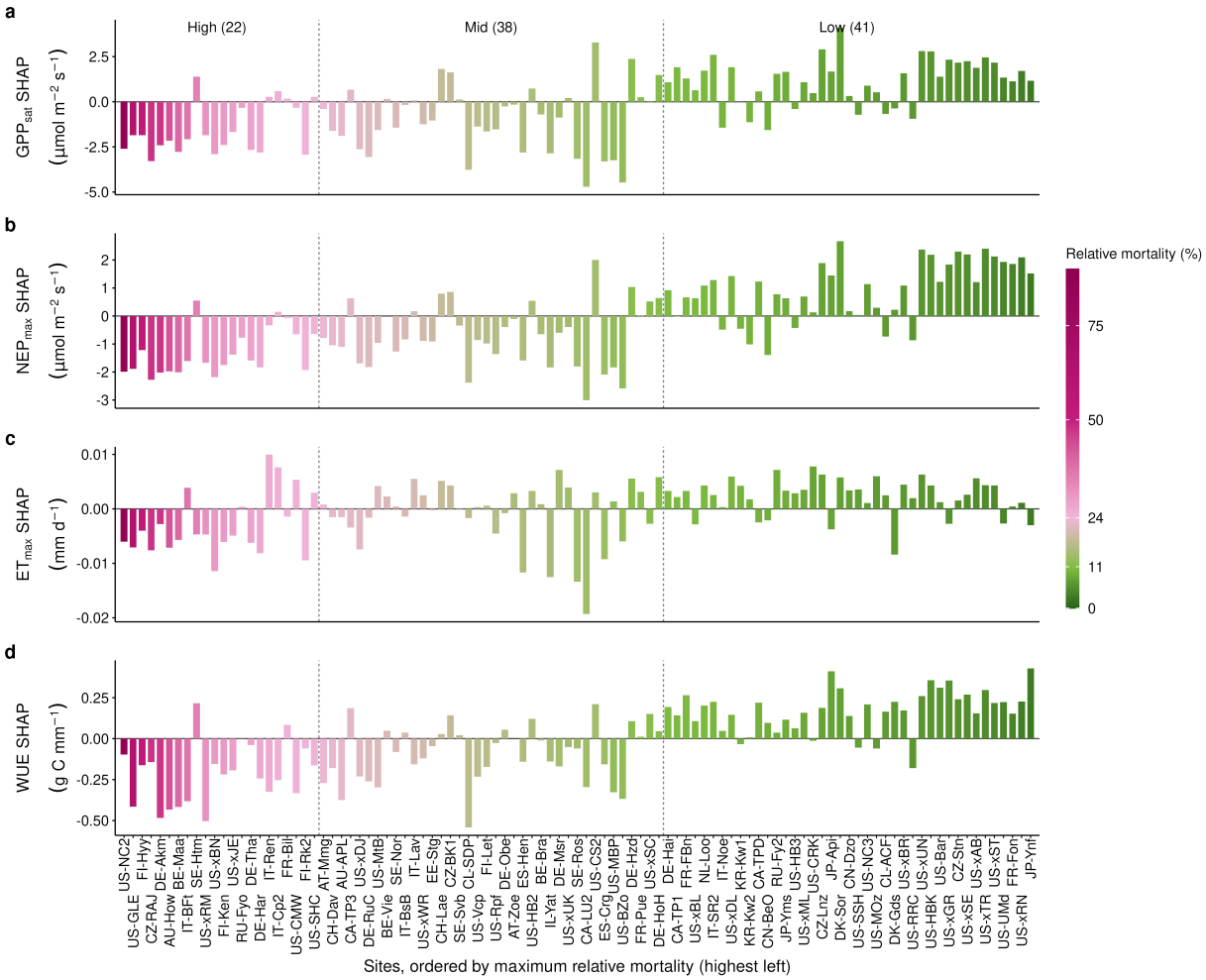


Fig. A7 | As Fig. 3 for XGBoost (Optuna-tuned, 24 months).

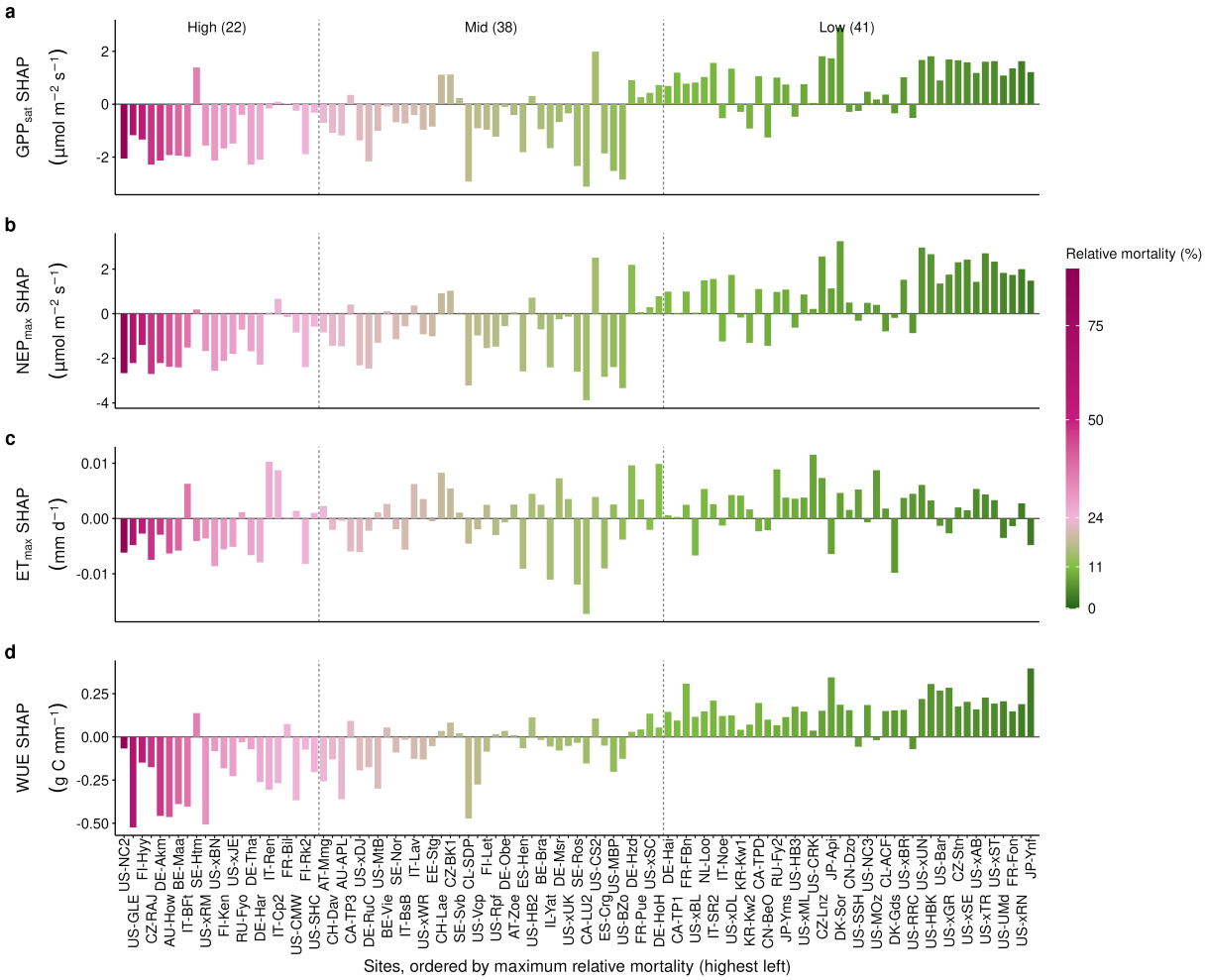


Fig. A8 | As Fig. 3 for LightGBM (Optuna-tuned, 24 months).

A4. High vs Low+Mid: all learners and relative deadwood

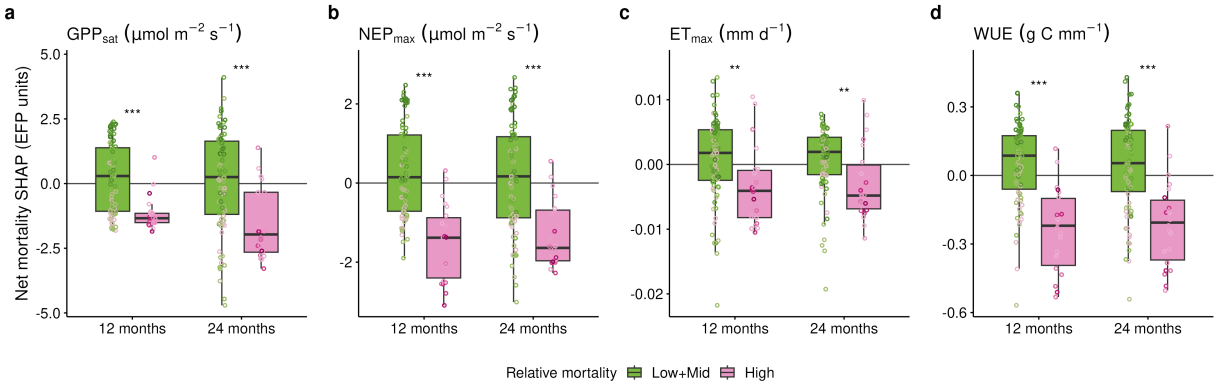


Fig. A9 | As Fig. 4 for XGBoost (Optuna-tuned).

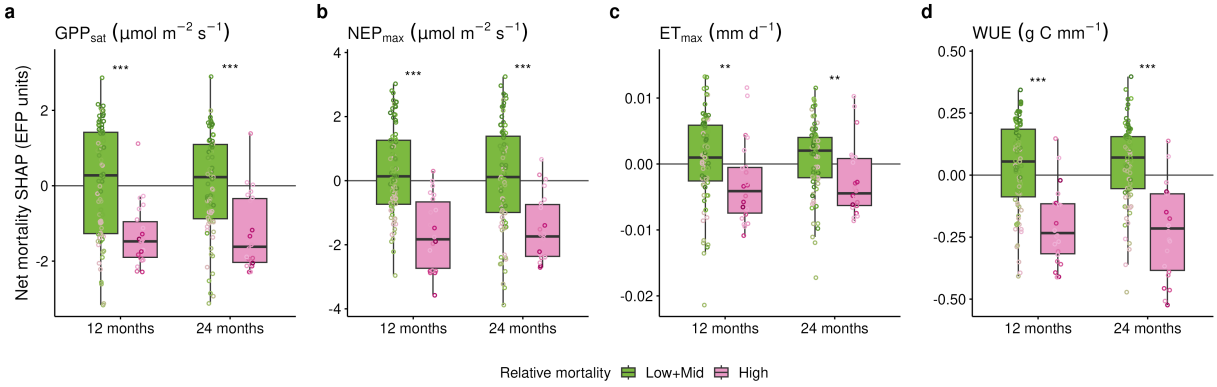


Fig. A9b | As Fig. 4 for LightGBM (Optuna-tuned).

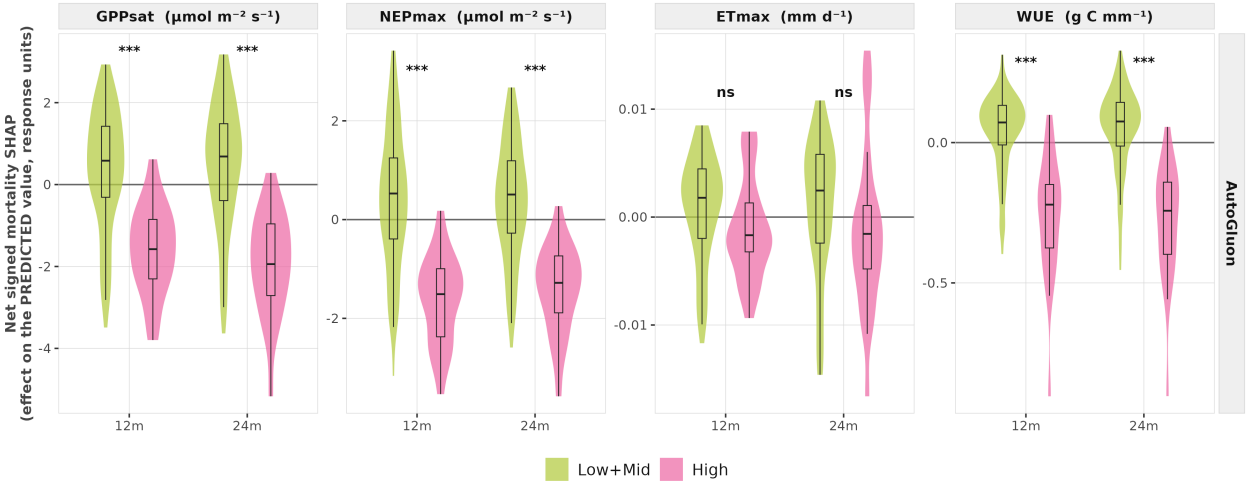


Fig. A10 | As Fig. 4 with the sites grouped by relative deadwood instead of relative mortality (AutoGluon). The High versus Low+Mid difference is the same for GPP_{sat}, NEP_{max} and WUE and not significant for ET_{max}.

A5. Ecosystem type and tree cover

- **A** ΔRMSE by IGBP, **B** net signed mortality SHAP by IGBP, **C** tree cover vs benefit, **D** tree cover vs direction, **E** relative mortality vs direction (= Fig. 5).
- C: no relationship (ρ = 0.10–0.18, n.s.). D: positive, ρ = +0.66 to +0.75 (all $p < 0.001$) — at low tree cover mortality pulls predicted carbon uptake down, at high tree cover it pushes it up.

- B: DBF, EBF and MF mostly positive, CSH, OSH and WET negative, ENF slightly negative. OSH has 2 sites, WSA 1, CSH and EBF 3 each, WET 6.

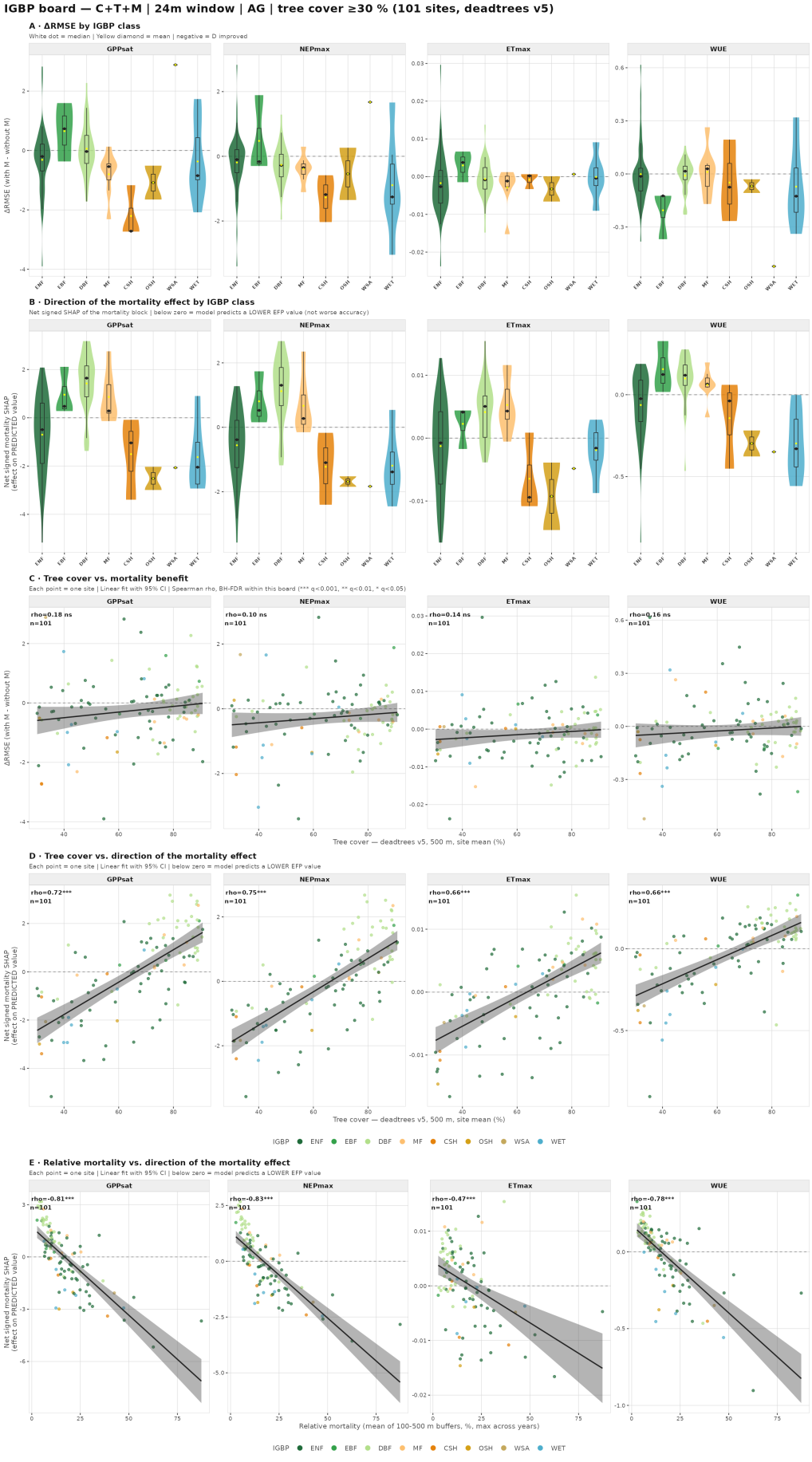

IGBP board — C+T+M | 24m window | AG | tree cover ≥30 % (101 sites, deadtrees v5)
A · ΔRMSE by IGBP class
White dot = median | Yellow diamond = mean | negative = D improved
GPPsat
NEPmax
ETmax
WUE
ΔRMSE (with M - without M)
B · Direction of the mortality effect by IGBP class
Net signed SHAP of the mortality block | below zero = model predicts a LOWER EFP value (not worse accuracy)
Net signed mortality SHAP (effect on PREDICTED value)
C · Tree cover vs. mortality benefit
Each point = one site | Linear fit with 95% CI | Spearman rho, BH-FDR within this board (*** q<0.001, ** q<0.01, * q<0.05)
rho=0.18 ns
n=101
rho=0.10 ns
n=101
rho=0.14 ns
n=101
rho=0.16 ns
n=101
Tree cover — deadtrees v5, 500 m, site mean (%)
D · Tree cover vs. direction of the mortality effect
Each point = one site | Linear fit with 95% CI | below zero = model predicts a LOWER EFP value
rho=0.72***
n=101
rho=0.75***
n=101
rho=0.66***
n=101
rho=0.66***
n=101
Tree cover — deadtrees v5, 500 m, site mean (%)
IGBP ENF EBF DBF MF CSH OSH WSA WET
E · Relative mortality vs. direction of the mortality effect
Each point = one site | Linear fit with 95% CI | below zero = model predicts a LOWER EFP value
rho=-0.81***
n=101
rho=-0.83***
n=101
rho=-0.47***
n=101
rho=-0.78***
n=101
Relative mortality (mean of 100-500 m buffers, %, max across years)
IGBP ENF EBF DBF MF CSH OSH WSA WET

Fig. A11 | The mortality effect by vegetation class and tree cover (AutoGluon, C+T+M, 24 months). **A**, change in RMSE when mortality is added, per IGBP class; **B**, net mortality SHAP per class; **C**, tree cover against the change in RMSE; **D**, tree cover against net mortality SHAP; **E**, relative mortality against net mortality SHAP (as Fig. 5). Each point is a site; lines, linear fits with 95 % confidence bands; ρ, Spearman rank correlation, Benjamini–Hochberg corrected within the figure.

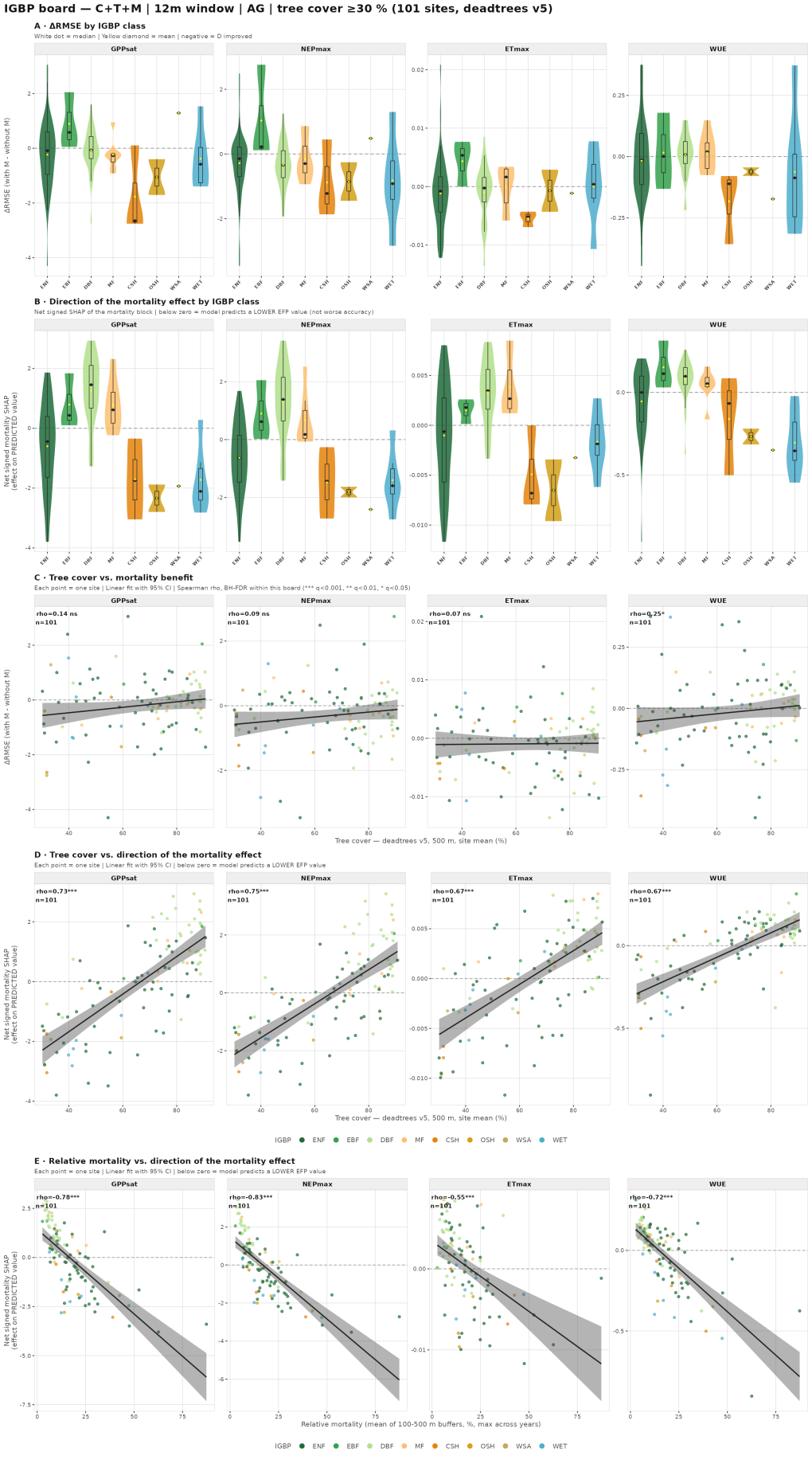


Fig. A12 | As Fig. A11 for the 12-month window (AutoGluon).

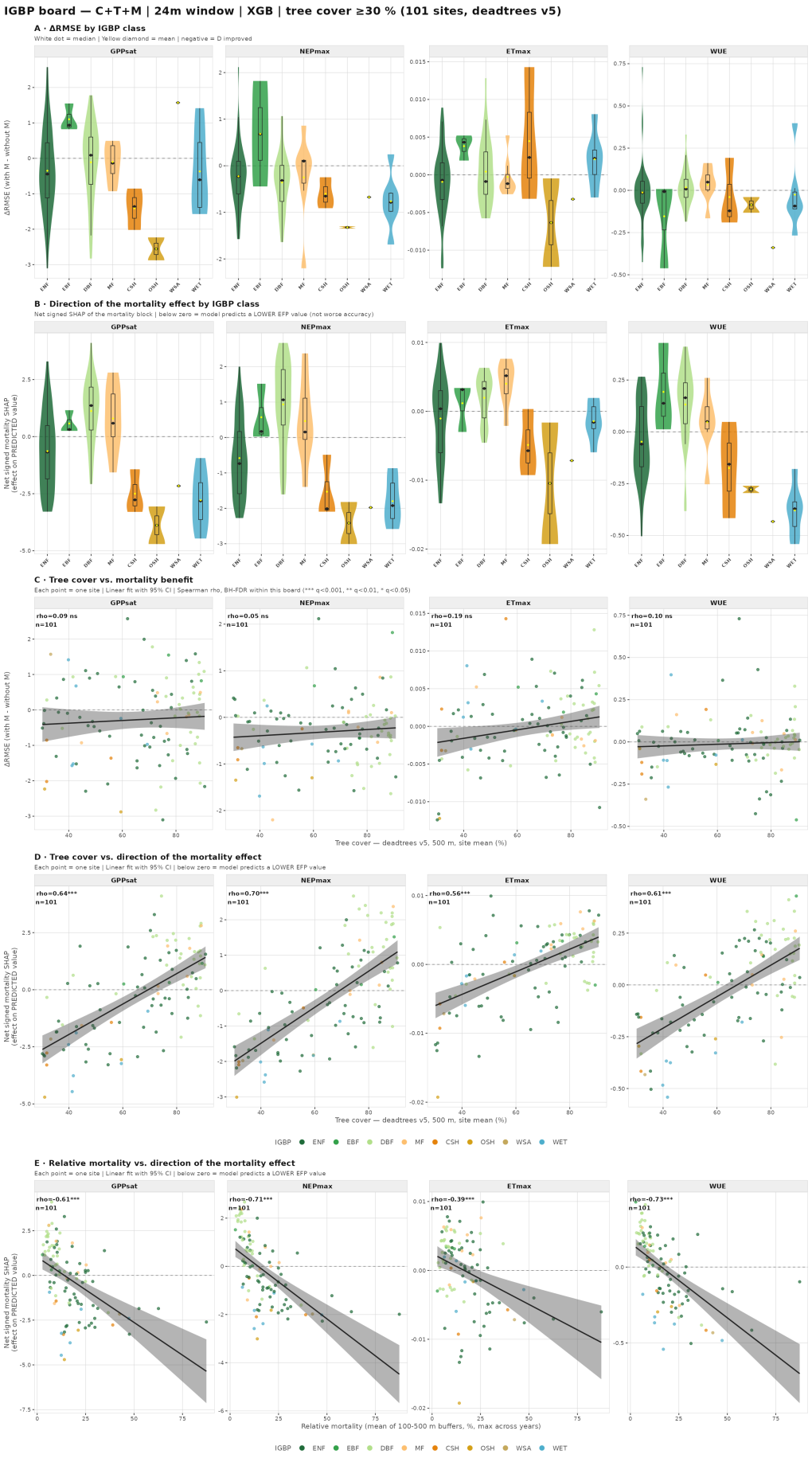


Fig. A13 | As Fig. A11 for XGBoost (Optuna-tuned, 24 months).

A6. Mortality metrics

- Share of the mortality block's mean |SHAP| per metric, AutoGluon: relative mortality 26 %, relative deadwood 24 %, relative tree-cover loss 22 %, absolute deadwood 16 %, deadwood gain 12 %; for ET_{max} tree-cover loss leads (32 %).
- Drop-column test (XGBoost and LightGBM retrained without each metric, or with only one metric added to C+T): removing any single metric changes the error by at most about 1 %, because the correlated metrics take over (relative mortality vs relative deadwood $\rho = 0.91$). Absolute deadwood gets 14–16 % of the SHAP but adds nothing when removed.

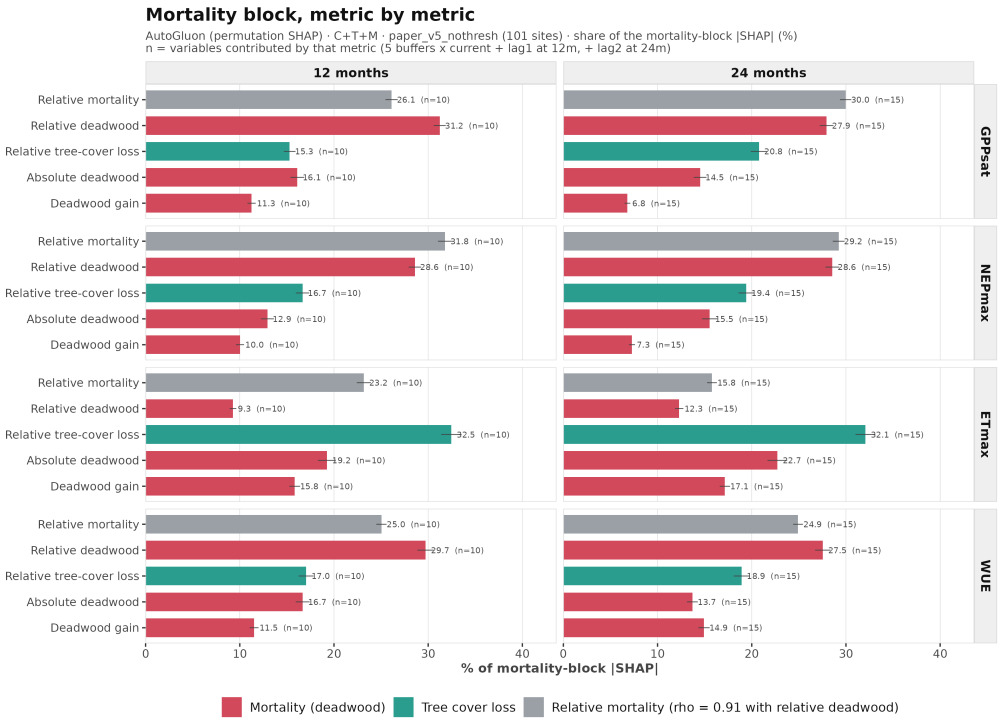


Fig. A14 | Contribution of each mortality metric to the mortality block (AutoGluon). Share of the block's mean |SHAP| per metric in the C+T+M model; bars, mean over sites ± s.e.

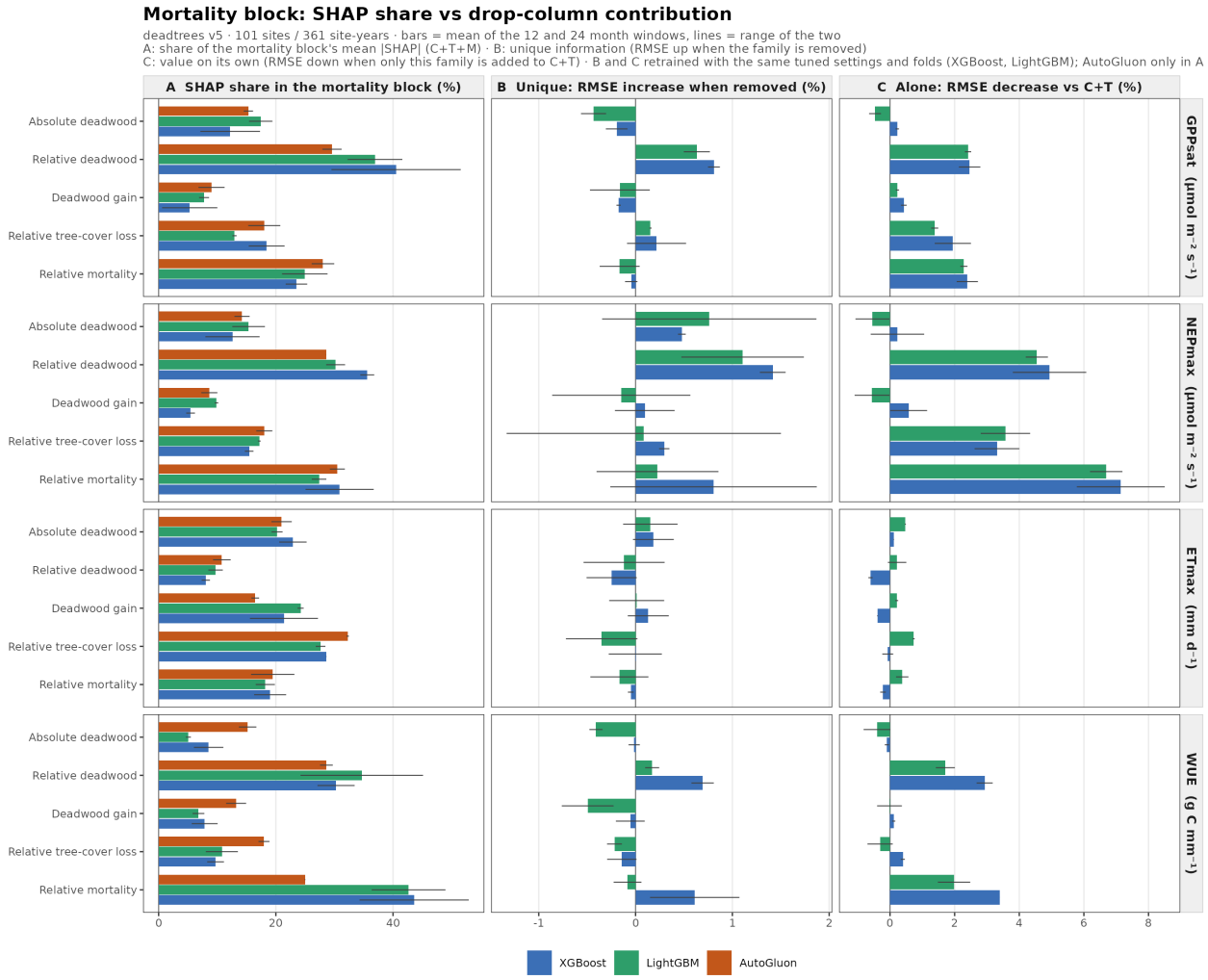


Fig. A15 | Removing or adding a single mortality metric changes the error little. **A**, share of each metric in the mortality block's |SHAP| (all learners); **B**, increase in RMSE when the metric is removed from C+T+M; **C**, decrease in RMSE when only this metric is added to C+T (**B** and **C**: XGBoost and LightGBM retrained with the same tuned settings and folds).

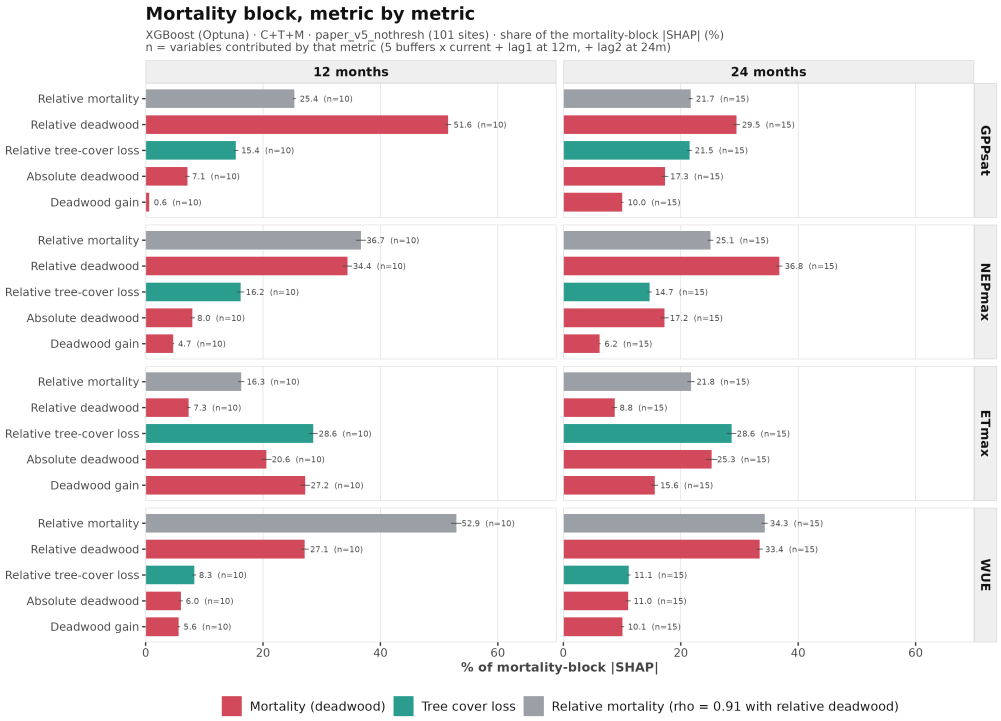


Fig. A16 | As Fig. A14 for XGBoost.

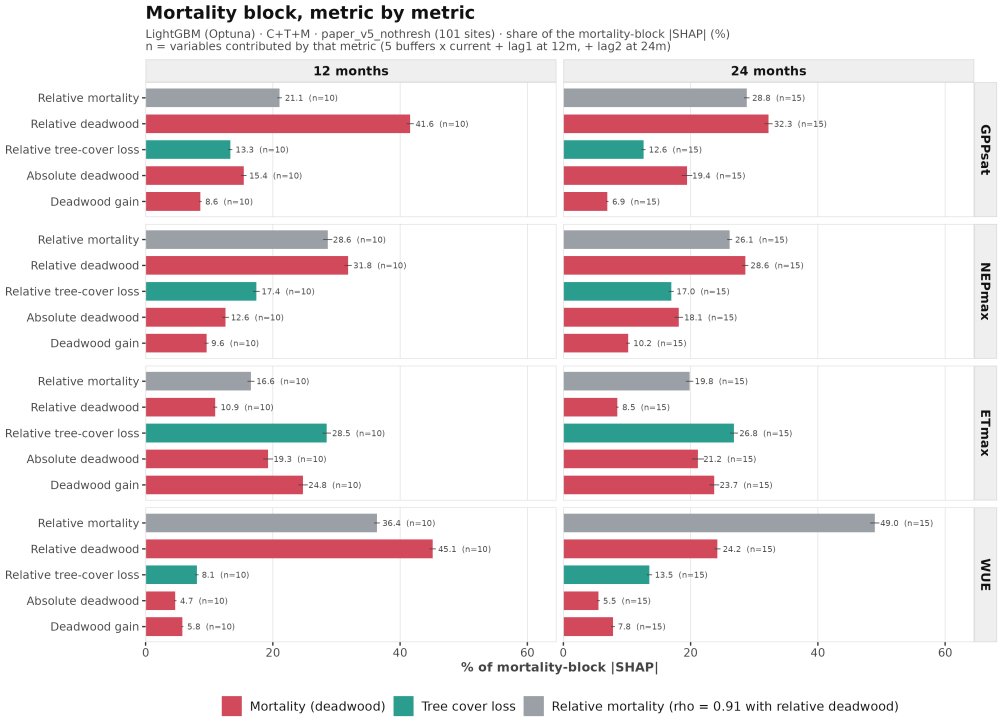


Fig. A17 | As Fig. A14 for LightGBM.

A7. Fig. 2: other learners, 12 months, mixed models and robustness

- XGBoost and LightGBM over-predict in the >30 % class as much as AutoGluon without mortality (NEP$_{max}$ −3.6 to −4.0 μmol m^{-2} s^{-1}); with mortality the over-prediction is smaller but stays below 0 (−1.8 to −2.4). Their correction in the >30 % class is significant for GPP$_{sat}$ (+1.4 to +1.8), NEP$_{max}$ (+1.2 to +2.2) and WUE (+0.12 to +0.19), all $q < 0.05$.

- Robustness of the mixed-model slopes (Fig. A21): mixed model, site-block bootstrap (1000 draws), leave one site out (range over 101 refits) and without the three most extreme sites (FI-

Hyy, US-GLE, US-NC2). The over-prediction without mortality (middle row) holds in every check for NEP_{max} and WUE; with mortality (bottom row) no check is significant for GPP_{sat}, NEP_{max} or WUE. Whether the error reduction itself grows with mortality (top row) goes the same way but is not robust in the bootstrap.

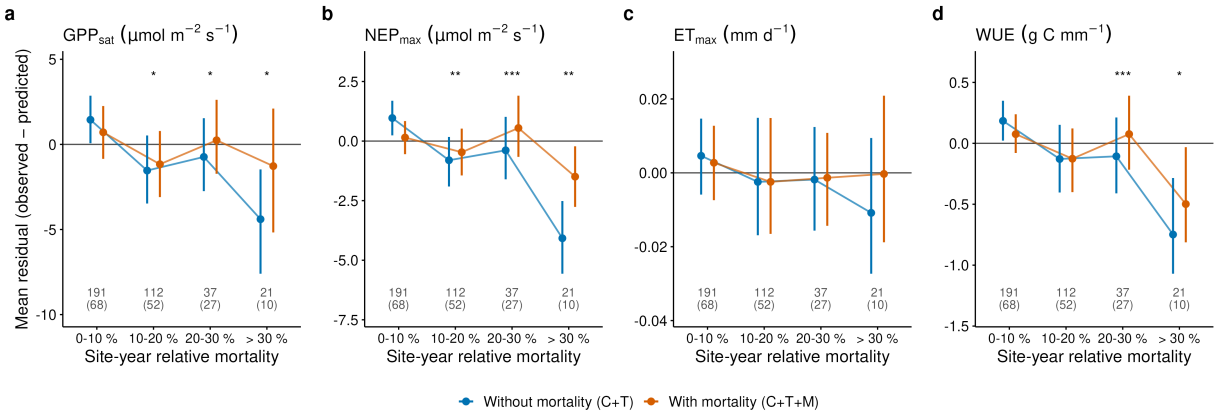


Fig. A18 | As Fig. 2 for the 12-month window (AutoGluon).

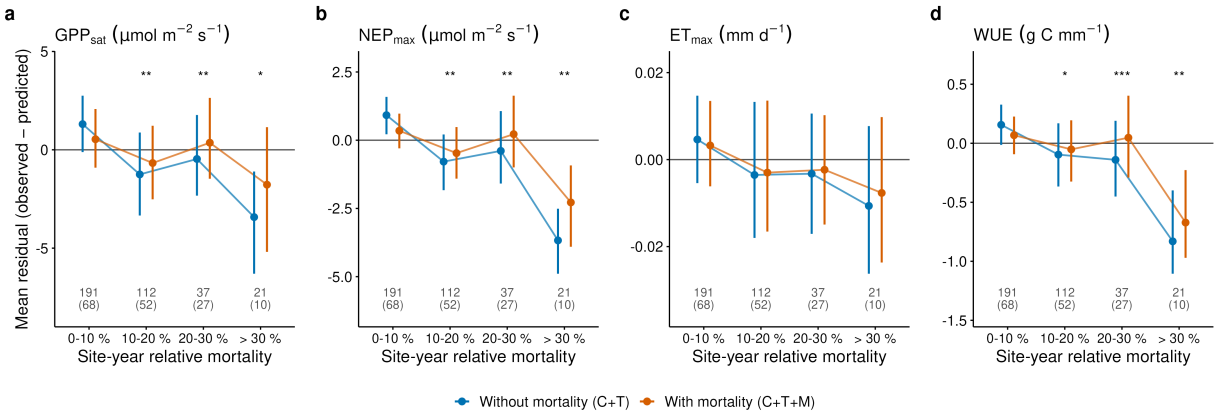


Fig. A19 | As Fig. 2 for XGBoost (Optuna-tuned, 24 months).

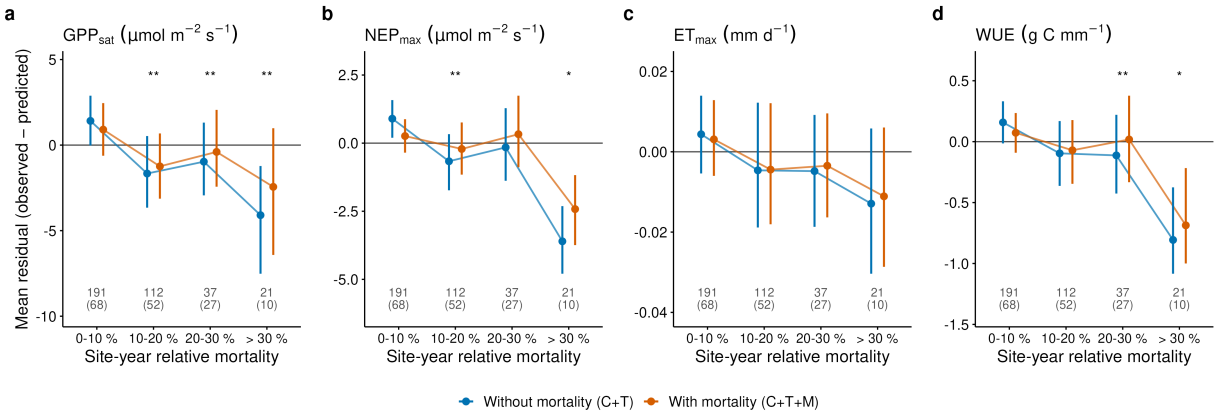


Fig. A20 | As Fig. 2 for LightGBM (Optuna-tuned, 24 months).

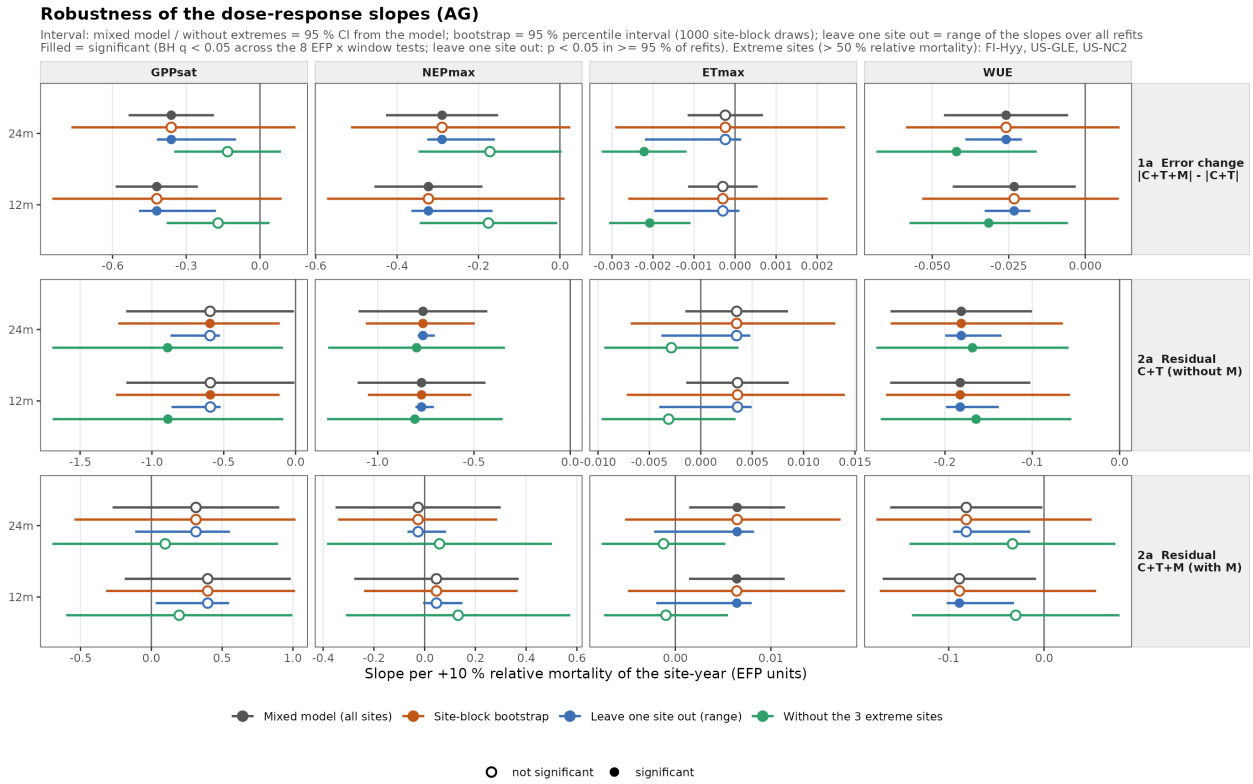


Fig. A21 | The over-prediction without mortality is robust (AutoGluon). Slope of the residual or error change per +10 % relative mortality of the site-year under each check: mixed model with a random intercept per site, site-block bootstrap (1,000 draws), leave one site out (range over 101 refits) and without the three sites with more than 50 % relative mortality. Filled symbols, significant.

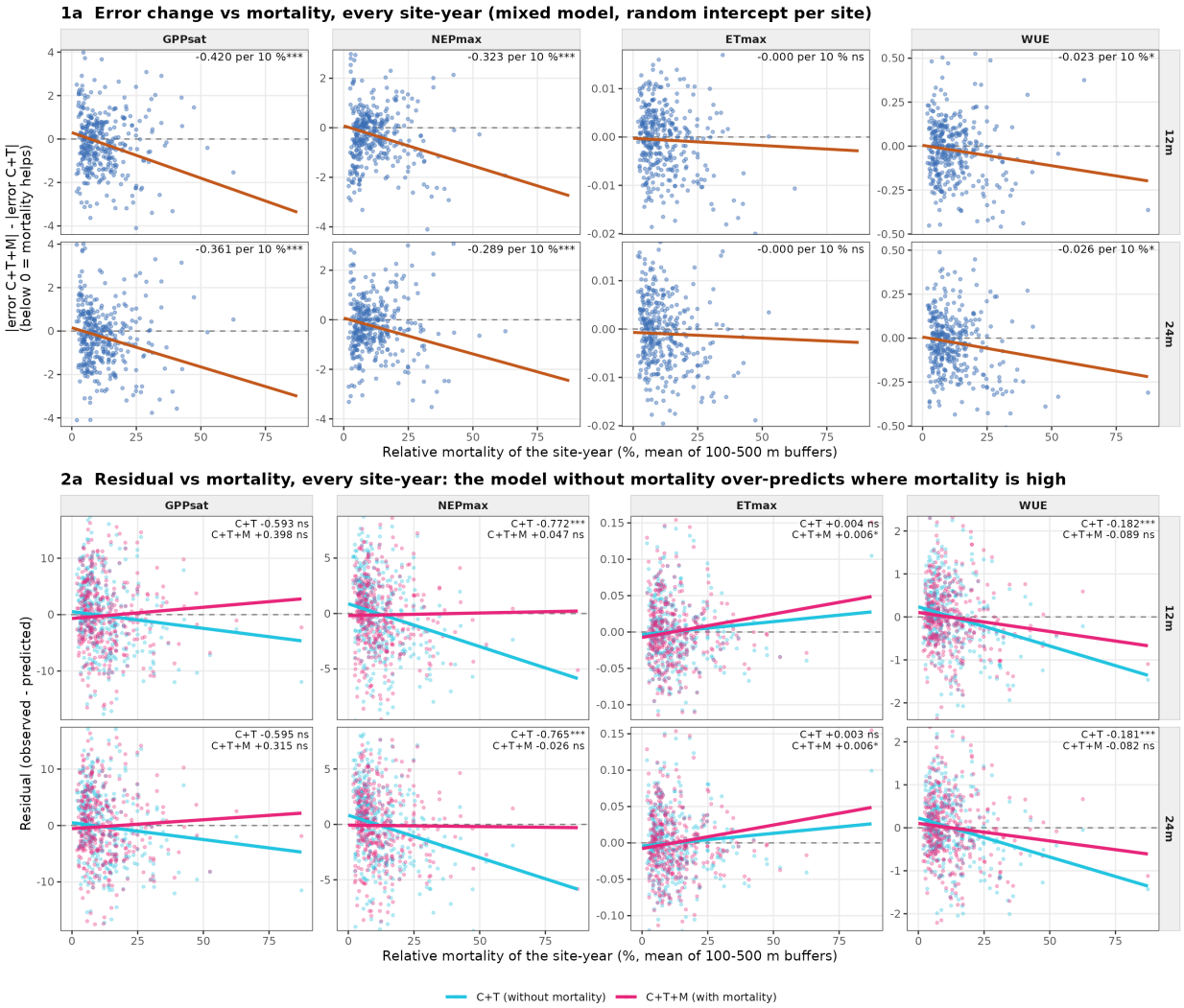


Fig. A22 | All site-years with the mixed-model slopes (AutoGluon). Top, change in error when mortality is added; bottom, residuals without (cyan) and with mortality (pink), against the relative mortality of the site-year.

A8. Circularity check: EFP-specific climate block

- Concern (Miguel): a climate variable that is part of an EFP's formula could make the prediction partly circular. The climate predictors are already strictly pre-season, but to be safe each EFP was refitted (AutoGluon, same design) **without the climate variables in its formula**: GPP_{sat} without radiation (light-response curve on PPFD = SW_{IN} × 2.11), ET_{max} and WUE without air temperature ($\mathrm{ET} = \mathrm{LE} / \lambda(\mathrm{T}_a)$); NEP_{max} has no climate variable in its formula and is unchanged. Filters (SW_{IN} thresholds, rain) are not part of the formulas and were kept. 36 (12 m) and 72 (24 m) climate predictors removed.

- **Skill hardly changes:** R^2 of C+T+M (12 / 24 m) GPP_{sat} 0.385 / 0.365 (before 0.367 / 0.370), ET_{max} 0.409 / 0.420 (0.425 / 0.433), WUE 0.281 / 0.302 (0.312 / 0.326).

- **The mortality effect is unchanged** (Table 9): WUE still improves significantly at High sites (−11 / −15 %) and in high-mortality years (−17 / −20 %); GPP_{sat} −4 to −16 %, significant at 12 m (all site-years −6.1 %, Mid+High years −8.3 %) but not at 24 m; ET_{max} not significant.

- **The over-prediction without mortality and its correction remain** (Fig. A23): in the >30 % class GPP_{sat} is over-predicted by 4.6 and corrected by +3.6 $\mu\mathrm{mol\,m^{-2}\,s^{-1}}$ ($q = 0.04$; 12 m: +3.7, $q = 0.009$), WUE by 0.71 and corrected by +0.25 $\mathrm{g\,C\,mm^{-1}}$ ($q = 0.04$; 12 m $q = 0.002$);

NEP_{max} as before.

- **So:** the mortality results do not depend on climate variables that enter the EFP formulas — no circularity from the climate block.

Table 9: **Adding mortality reduces the prediction error (MAE), most in the years in which trees died.** Change in the MAE when the mortality predictors are added to climate and traits (C+T → C+T+M), AutoGluon and AutoGluon, EFP-specific climate, leave-one-site-out predictions: $100\,(\text{MAE}_{\text{C+T+M}} / \text{MAE}_{\text{C+T}} - 1)$ %; negative = smaller error. Brackets: 95 % interval of a bootstrap that resamples whole sites (10 000 draws). Stars: one-sided paired permutation test with the two models swapped per site (10 000 draws), Benjamini–Hochberg corrected across both runs: $^{*}\,q < 0.05$, $^{**}\,q < 0.01$, $^{***}\,q < 0.001$. *Site-year level*: every site-year is one observation. *All years of the sites*: all sites, or only the sites whose maximum relative mortality exceeds 10.6 % (Mid + High) or 24.0 % (High), with all their years. *Only the years with mortality*: site-years whose own relative mortality exceeds 10.6 % or 24.0 %. Numbers under the column heads: sites / site-years. 101 sites, 361 site-years, deadtrees v5 maps.

		All years of the sites			Only the years with mortality	
		All sites (101 / 361)	Mid + High sites (60 / 225)	High sites (22 / 80)	> 10.6 % (60 / 160)	> 24.0 % (22 / 40)
AutoGluon 12 months	GPP_{sat}	−4.1 [−8.0, −0.2]*	−6.1 [−10.9, −1.5]*	−7.2 [−18.1, +2.9]	−6.7 [−12.8, −0.7]*	−12.0 [−29.5, +4.9]
	NEP_{max}	−11.8 [−17.0, −6.3]**	−11.3 [−17.5, −5.1]**	−22.6 [−33.6, −11.1]**	−12.5 [−20.1, −4.9]**	−31.6 [−43.2, −16.0]*
	ET_{max}	−1.3 [−4.2, +1.3]	−2.1 [−5.3, +1.1]	−2.9 [−8.4, +2.1]	−3.9 [−7.7, +0.0]*	−9.7 [−20.6, +2.7]
	WUE	−4.3 [−8.1, −0.2]*	−6.8 [−11.3, −2.3]**	−10.9 [−17.8, −4.3]**	−7.2 [−12.2, −1.8]*	−16.9 [−25.2, −5.1]*
AutoGluon 24 months	GPP_{sat}	−5.7 [−9.5, −1.7]*	−8.3 [−12.6, −4.0]**	−7.1 [−17.2, +3.6]	−8.3 [−14.2, −2.6]*	−11.1 [−28.0, +8.1]
	NEP_{max}	−10.8 [−16.0, −5.1]**	−10.5 [−17.3, −3.4]*	−17.5 [−30.7, −2.1]*	−11.6 [−19.8, −2.7]*	−29.2 [−42.9, −8.6]*
	ET_{max}	−2.0 [−5.0, +0.9]	−3.7 [−7.3, −0.1]*	−5.3 [−13.1, +1.2]	−5.4 [−9.6, −1.3]*	−11.4 [−23.7, +3.2]
	WUE	−4.7 [−9.1, −0.1]*	−6.4 [−11.5, −1.2]*	−14.8 [−23.9, −6.4]*	−8.2 [−14.5, −1.5]*	−21.8 [−34.5, −8.2]*
AutoGluon, EFP-specific climate 12 months	GPP_{sat}	−6.1 [−10.5, −1.7]*	−7.8 [−13.7, −1.9]*	−12.1 [−24.4, +0.5]	−8.3 [−15.6, −1.1]*	−16.1 [−35.0, +5.0]
	NEP_{max}	−11.8 [−17.0, −6.3]**	−11.3 [−17.5, −5.1]**	−22.6 [−33.6, −11.1]**	−12.5 [−20.1, −4.9]**	−31.6 [−43.2, −16.0]*
	ET_{max}	−1.5 [−4.4, +1.4]	−1.8 [−5.0, +1.8]	+0.4 [−5.5, +8.0]	−2.2 [−5.9, +2.4]	−2.0 [−17.0, +14.8]
	WUE	−4.3 [−7.9, −0.1]*	−6.6 [−11.1, −1.9]*	−11.2 [−18.6, −4.4]*	−6.3 [−12.0, −0.2]*	−16.8 [−26.6, −5.5]*
AutoGluon, EFP-specific climate 24 months	GPP_{sat}	−4.1 [−8.4, +0.2]	−4.0 [−10.2, +2.0]	−4.5 [−19.0, +11.6]	−4.7 [−12.6, +3.2]	−9.9 [−33.3, +19.7]
	NEP_{max}	−10.8 [−16.0, −5.1]**	−10.5 [−17.3, −3.4]*	−17.5 [−30.7, −2.1]*	−11.6 [−19.8, −2.7]*	−29.2 [−42.9, −8.6]*
	ET_{max}	−1.1 [−4.1, +1.7]	−1.8 [−5.4, +1.7]	−1.1 [−8.3, +6.1]	−3.5 [−8.0, +1.1]	−6.3 [−20.7, +9.3]
	WUE	−4.2 [−8.5, +0.5]	−5.7 [−11.1, −0.0]*	−14.5 [−22.5, −6.6]**	−7.9 [−14.0, −1.2]*	−20.4 [−29.2, −9.5]*

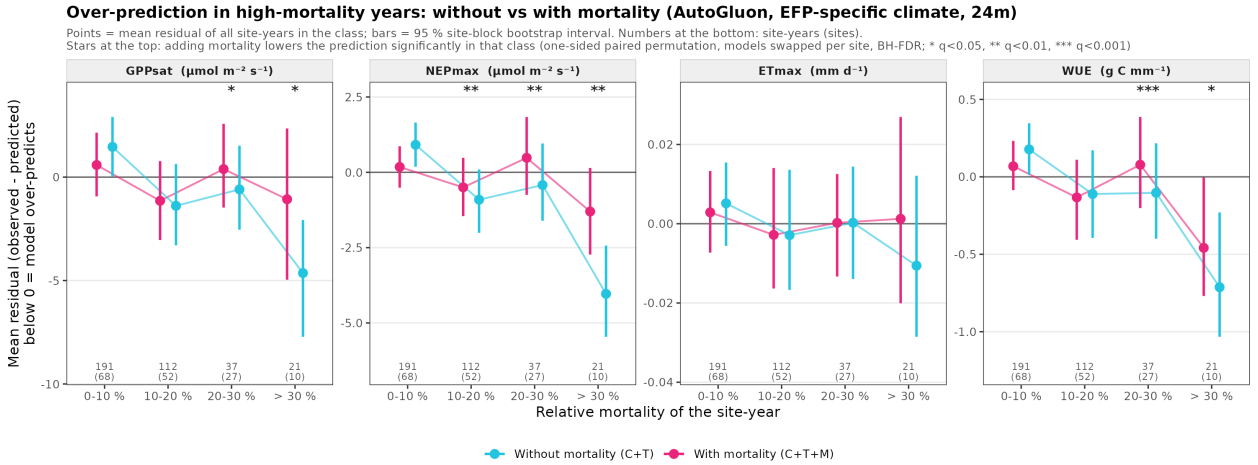


Fig. A23 | The over-prediction and its correction remain when the climate variables used to compute each property are left out. As Fig. 2 (AutoGluon, 24 months) with the EFP-specific climate block: radiation removed for GPP_{sat}, air temperature for ET_{max} and WUE.